Informal learning in everyday human–LLM interaction

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Abstract

As LLM-based assistants become embedded in everyday work and study, lifelong learning is increasingly mediated by everyday human–LLM interaction. Yet it remains unclear whether these interactions elicit the engagement through which people build understanding. We analyse 128,334 naturalistic conversations from three public conversational corpora—WildChat, LMSYS Chat and ShareChat—adapting learning-science constructs to 979,974 turns through human-verified LLM-assisted annotation. Across 490,932 user turns, cognitive engagement appeared in 31.9% and constructive engagement—the highest-level observable form in which users elaborate, test, revise or extend ideas—in 4.9%. Constructive engagement was organized by user framing, task ecology and sustained exchange across the three corpora. On the assistant side, support was most informative when analysed not as a binary scaffolding signal, but as a situated match between support form and user activity: feedback-like and explanatory forms aligned with constructive engagement, and adjacent-turn analyses showed local, state- and form-dependent coupling rather than a demonstrated global trajectory. These findings make informal learning in everyday AI use measurable and show how AI-mediated work can preserve opportunities for people to build, test and extend understanding while solving real problems.

Keywords: human–LLM interaction, informal learning, large language models, scaffolding, learner engagement

1 Introduction

Large language models (LLMs) are becoming embedded in the cognitive work of everyday life. People use them to write, code, search, troubleshoot and make sense of unfamiliar problems, often outside classrooms, training programmes or purpose-built tutoring systems. In such moments, the model may do more than help complete the task at hand: it may shape what the user notices, understands and tries next. Everyday human–LLM interaction therefore constitutes a rapidly expanding setting for informal learning, in which understanding can emerge through everyday activity, self-directed inquiry and problem solving rather than formal curricula [1–3]. Yet existing research on LLMs and learning has focused primarily on settings where learning support is deliberately designed, such as automated feedback, question generation, retrieval practice and tutoring systems [4–8]. These studies show that LLMs can enact instructional strategies when explicitly prompted, trained or embedded in purpose-built systems [9, 10]; they tell us less about whether general-purpose LLMs used for everyday tasks actually keep users engaged in the work of understanding. The educational significance of everyday AI use therefore depends on the interactional dynamics between model support and user activity: whether users ask questions, test assumptions, revise ideas and extend solutions, or whether AI-mediated work simply makes answer delivery faster while leaving human understanding opaque. We ask how often users show learning-oriented engagement in everyday human–LLM interaction, and which contextual and interactional conditions make it more likely.

We analyse 128,334 naturalistic conversations from three public conversational corpora: WildChat, LMSYS Chat and ShareChat [11–13]. Across these corpora, we focus on coding and writing as two common task ecologies of everyday LLM use. The corpus contains 490,932 user turns and 489,042 assistant turns. The analysis proceeds in three layers: first, whether learning-oriented engagement is visible and how it is composed; second, where constructive engagement is concentrated across user framing, task ecology and interaction depth; and third, how assistant scaffolding is associated with constructive participation across conversation-level, support-form and adjacent-turn temporal scales. To make these questions measurable in logs without fixed instructional tasks, explicit learning objectives or post-test outcomes, we translate theories of engagement and scaffolding into role-specific, turn-level behavioural signatures: user turns are coded for passive, active or constructive uptake [14, 15], while assistant turns are coded for the presence of scaffolded support and further characterized by support intent and support form [16–19]. Following the literature that deeper forms of overt engagement are more strongly linked to learning [15], we treat constructive engagement as the highest-level observable indicator of learning-oriented participation available in naturalistic logs. These measures do not establish causal learning effects or durable knowledge gains; instead, they allow us to observe whether learning-oriented engagement is present in everyday use and to test which contextual and interactional conditions are associated with deeper user participation.

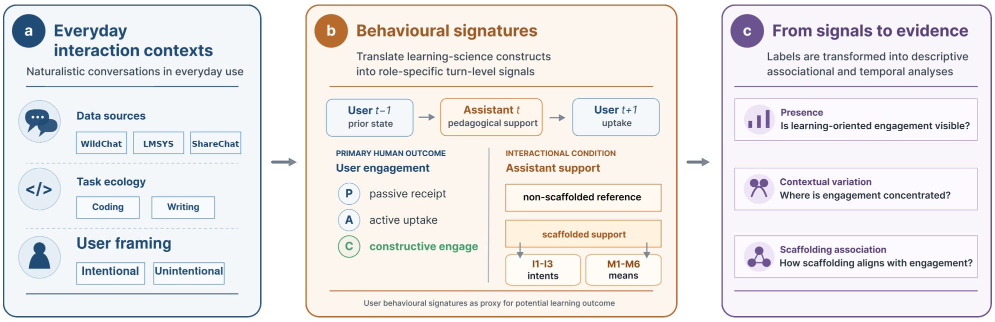


Fig. 1 Analytic framework for studying informal learning in everyday human–LLM interaction. a, Naturalistic public conversations are situated by task ecology and user framing. **b,** Learning-science constructs are translated into turn-level behavioural signatures for user engagement and assistant support. **c,** The resulting labels support analyses of engagement presence, contextual variation and scaffolding–engagement association. Labels capture observable interactional signatures, not durable learning outcomes or causal instructional effects.

We find, first, that learning-oriented engagement is a recurrent feature of everyday human–LLM interaction, although its constructive form is selective. Across the three public corpora, cognitive engagement appears in 31.9% of user turns and constructive engagement in 4.9%, while emotional engagement is rare; scaffolded support appears in 31.7% of assistant turns. These rates show that everyday AI use often involves more than answer delivery, but the deepest form of user engagement is a less frequent behavioural signal. Second, constructive engagement is organized by context: it is amplified by intentional user framing, varies across task ecologies and becomes most visible in sustained exchanges where users return to, interrogate and reshape the assistant's response. Third, assistant scaffolding marks richer constructive participation across the six task settings, but not as a uniform or treatment-like effect. Its association with engagement depends on user framing, support form, task-specific support supply and temporal scale: feedback-like and explanatory support align most clearly with constructive engagement, whereas more directive or questioning forms are less aligned with visible elaboration, and the strongest temporal evidence is local, state-dependent and form-dependent rather than evidence for a uniform global trajectory. Together, these findings locate everyday human–LLM interaction in a middle space between ordinary tool use and designed instruction. By making this middle space observable, the study frames a human-centred challenge for AI-mediated work: preserving opportunities for people to build, test and extend understanding while solving real problems, without turning every interaction into a formal lesson.

2 Results

The results proceed from engagement presence to the conditions that differentiate its depth. We first ask whether everyday human–LLM conversations contain learning-oriented engagement, distinguishing broad cognitive uptake from the highest-level constructive form. We then examine where constructive engagement is concentrated

Table 1 Dataset and behavioural overview. Summary statistics for the WildChat, LMSYS Chat and ShareChat task settings analysed in the study. Constructive engagement is reported as the highest-level form within cognitive engagement and serves as the focal user-side outcome in subsequent analyses.

	Sample size			User framing	Assistant support	Cognitive engagement		Emotional engagement
Setting	Conversations	User turns	Assistant turns	Intentional conv. (%)	Scaffolded turns (%)	Overall (%)	Constructive level (%)	Emotional (%)
WildChat coding	31,878	124,073	124,623	37.13	51.03	50.21	9.23	0.95
WildChat writing	39,534	163,705	164,824	25.80	23.53	16.34	2.22	0.57
LMSYS coding	31,879	106,312	104,431	41.54	29.11	45.28	5.43	0.90
LMSYS writing	21,023	77,906	76,279	20.64	19.43	15.30	1.53	0.21
ShareChat coding	2,481	11,541	11,522	45.63	50.67	45.00	15.17	3.38
ShareChat writing	1,539	7,395	7,363	29.69	22.42	33.05	5.18	0.34
Total / pooled	128,334	490,932	489,042	32.11	31.71	31.93	4.93	0.74

Note. Percentages for cognitive, constructive and emotional engagement are computed over user turns; percentages for scaffolded support are computed over assistant turns. The constructive-level column reports the share of all user turns that reached the constructive depth level within cognitive engagement. Figure 2a reports the composition of passive, active and constructive uptake among cognitively engaged turns.

across user framing, task ecology and sustained exchange. Finally, we ask how assistant scaffolding relates to constructive engagement: first as a conversation-level condition, then through support forms, and finally through adjacent-turn temporal ordering. Because the logs contain no direct assessment of knowledge gain, engagement labels are interpreted as behavioural signatures of potential learning, and scaffolding analyses are interpreted as associations rather than causal evidence of learning effects.

2.1 Everyday human–LLM conversations contain learning-oriented engagement

Across the six WildChat, LMSYS Chat and ShareChat task settings, 31.9% of user turns showed cognitive engagement, 4.9% reached constructive engagement, and emotional engagement was rare, appearing in only 0.74% of user turns (Table 1). This establishes the first descriptive pattern: users often engage with assistant responses in learning-oriented ways rather than merely passively acknowledging or receiving task output, while their affective engagement is limited.

Figure 2a decomposes cognitively engaged turns into passive receipt, active uptake and constructive engagement across the six task settings. Active uptake was the dominant form of cognitive engagement, while the highest-level constructive form made up a smaller but stable subset. A majority of cognitively engaged turns involved active uptake: users used, applied or followed up on the assistant's response. A smaller but stable subset involved constructive engagement, in which users elaborated, tested, revised or extended ideas. This distinction matters for the rest of the Results. Cognitive engagement provides the broad signal that everyday LLM use can become learning-oriented, active uptake shows that users often work with the assistant's response, and constructive engagement identifies the deepest cases of user participation that we seek to explain. The following analyses therefore focus on where this constructive form appears and which interactional conditions are associated with it.

2.2 Constructive engagement is shaped by user framing, task ecology and interaction depth

Having established that everyday human–LLM conversations contain learning-oriented engagement, we next examined where its constructive form was concentrated. User framing provided the first contrast (Fig. 2b). Across the six main settings, conversations explicitly oriented toward learning or exploration showed higher engagement than conversations without such framing. Importantly, both explicitly learning-oriented and more task-oriented conversations contained constructive participation, indicating that informal learning-relevant engagement was not confined to interactions that announced themselves as learning episodes.

Task ecology provided a second source of variation. In the two everyday task ecologies sampled here, coding-oriented conversations showed higher constructive engagement than writing-oriented conversations across WildChat, LMSYS and ShareChat. Cognitive engagement was 45.00–50.21% in coding conversations, compared with 15.30–33.05% in writing conversations; constructive engagement was 5.43–15.17% in coding and 1.53–5.18% in writing. The LMSYS coding setting illustrates why broad cognitive uptake and deeper constructive engagement should remain separate outcomes: because LMSYS Chat reflects an earlier public-demo and Chatbot Arena period [12], many coding exchanges involved active repair questions, added constraints and requests for another direct coding step, which can raise active uptake without producing the same level of constructive elaboration. One plausible reason for a higher engagement rate of coding tasks is that they often make uncertainty and failure externally visible through errors, constraints and executable tests, whereas writing tasks may more often be resolved through drafting, editing or applying a proposed revision without requiring users to articulate the underlying rationale [20–22]. This pattern suggests that task nature organized the baseline opportunities for users to ask questions, test assumptions, refine solutions and extend ideas.

Interaction depth provided a third lens on where constructive participation surfaced (Fig. 2c). Across the six main settings, the probability that a conversation contained at least one constructive user turn increased with behavioural depth. Because this measure asks whether a rare event occurred at least once, part of the gradient is mechanical: longer conversations provide more opportunities to observe a constructive turn. We therefore interpret Fig. 2c as a descriptive visibility pattern, not as evidence that length itself produces deeper engagement. A grouped-binomial rate sensitivity, modelling constructive user-turn counts out of total user turns with user framing, task ecology, length bucket and dataset fixed effects, showed a much smaller but still positive length association (OR=1.13 for 4–6 user turns; OR=1.06 for 7+ user turns; Supplementary Table C9). The substantive point is that constructive engagement is most visible when everyday LLM use becomes a sustained exchange rather than a one-off request for an answer.

Together, these results show that constructive engagement is ecological. It is amplified by intentional user framing, varies across task ecologies and is most visible in sustained exchanges. A simple conversation-level logistic check showed the same directional organization after including user framing, task ecology, conversation-length

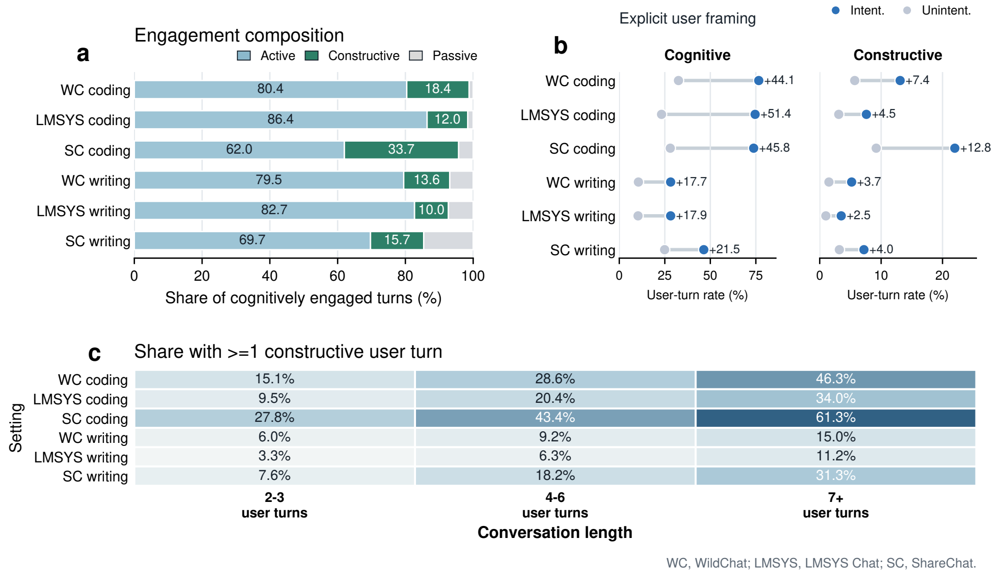


Fig. 2 Learning-oriented engagement varies by engagement form, user framing and conversation length. a, Composition of cognitively engaged user turns by active, constructive and passive levels across the six task settings. **b,** Cognitive and constructive engagement rates in conversations with intentional versus unintentional user framing. **c,** Share of conversations with at least one constructive user turn by conversation-length bucket. Panel a uses cognitively engaged turns as the denominator; panel b reports percentages of user turns; panel c reports conversation-level shares. WC, WildChat; LMSYS, LMSYS Chat; SC, ShareChat.

bucket and dataset fixed effects (Supplementary Table C8), and the constructive-rate sensitivity shows that the length pattern is not only an opportunity-count artifact.

2.3 Scaffolded conversations show richer constructive participation

After characterizing user-side conditions such as user framing and task ecology, we next examined how assistant behaviour was associated with richer constructive participation through scaffolding, a canonical way that teachers support engagement. At the conversation level, conversations containing at least one scaffolded assistant turn had higher constructive ratios than conversations without scaffolded support in all six task settings (Fig. 3a). Using the same turn-weighted estimator in Fig. 3a, Table 2 and Supplementary Table C6, these raw constructive-ratio differences ranged from +0.99 to +7.34 percentage points across WildChat, LMSYS and ShareChat, and scaffolded conversations also showed greater post-first-response depth in every setting (+1.43 to +3.38 turns; Fig. 3d). All six constructive-ratio contrasts and all six depth contrasts were statistically distinguishable from zero in conversation-level bootstrap tests ($p < .001$; Supplementary Table C6). These descriptive contrasts show that scaffolding co-occurred with both more constructive participation and more sustained exchange; the percentage-point differences are reported as effect sizes.

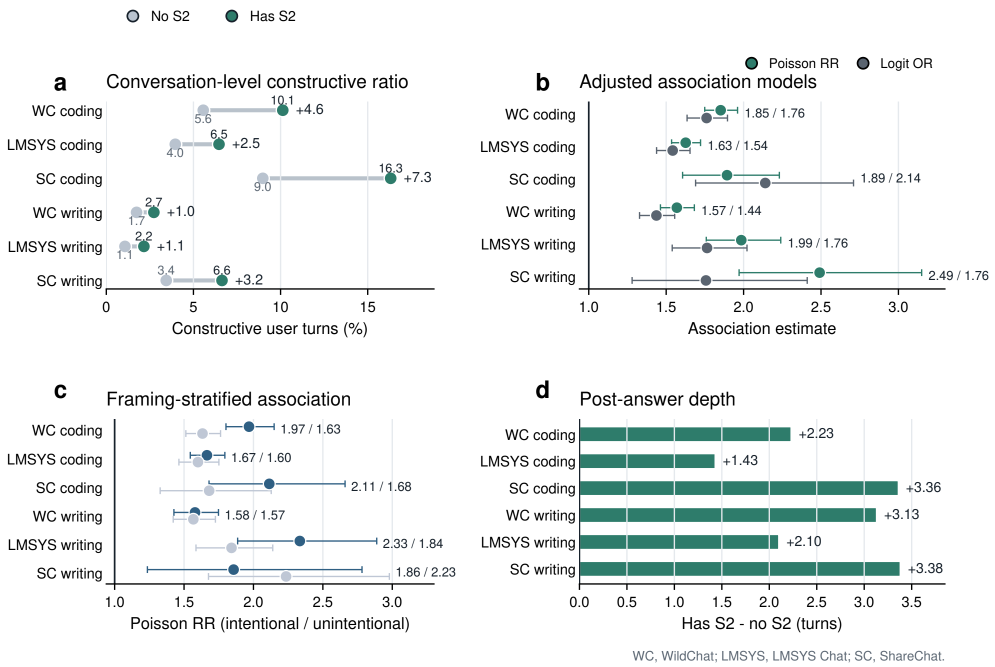


Fig. 3 Scaffolded conversations are associated with richer constructive participation across the three corpora. a, Turn-weighted constructive user-turn ratios in conversations with versus without scaffolded support. **b,** Covariate-adjusted associations between scaffolded-support presence and constructive participation, reported as Poisson count ratios and logistic odds ratios. **c,** Poisson count ratios for scaffolded-support presence stratified by intentional versus unintentional user framing. **d,** Difference in post-answer interaction depth after the first assistant response. Numbers indicate scaffolded-minus-reference differences, except panels b and c, which report model estimates. Error bars in panels b and c show model-based 95% confidence intervals; Poisson overdispersion and offset-rate sensitivities are reported in Methods and the Appendix.

The association remained positive after adjusting for observed user framing, task ecology and conversation-level context in covariate-adjusted models. Across the six settings, Poisson count ratios for constructive-turn counts ranged from 1.569 to 2.491, and logistic odds ratios for whether a conversation contained at least one constructive turn ranged from 1.437 to 2.140 (Fig. 3b). These count ratios and odds ratios are reported as model-based effect-size estimates, indicating that scaffolded support remained associated with higher constructive participation after observed-covariate adjustment. A setting-specific offset-rate sensitivity, using `log(user_turns)` as the exposure offset and quasi-Poisson scaled standard errors, preserved positive `has_S2` associations in all six settings (rate ratios 1.356–1.703; all $p < .001$; Supplementary Table C7).

Table 2 Summary of key support–engagement associations across the three corpora. Constructive-ratio contrasts compare conversations with versus without scaffolded support using turn-weighted constructive user-turn ratios. Adjusted models report the association of scaffolded-support presence with constructive-turn count (Poisson count ratio) and with having at least one constructive turn (logit odds ratio). Adjacent-turn lift is the difference in the probability that the next user turn is constructive after scaffolded versus non-scaffolded reference assistant turns.

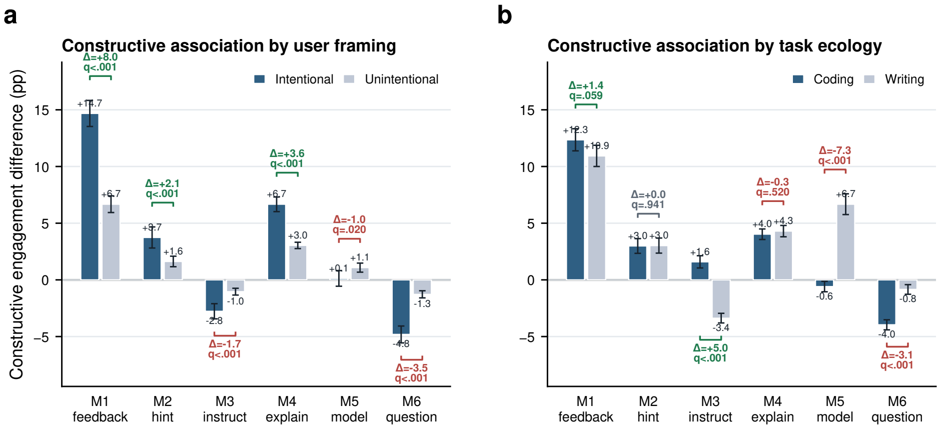
Setting	Constructive ratio Has S2 / No S2	Difference (pp)	Depth diff. (turns)	Poisson count ratio	Logit OR	Adjacent lift (pp)	Around first S2 diff.
WildChat coding	0.101 / 0.056	+4.6	+2.23	1.852	1.761	+2.68	+0.014
LMSYS coding	0.065 / 0.040	+2.5	+1.43	1.626	1.542	+2.01	+0.015
ShareChat coding	0.163 / 0.090	+7.3	+3.36	1.893	2.140	+6.21	+0.007
WildChat writing	0.027 / 0.017	+1.0	+3.13	1.569	1.437	+1.13	-0.042
LMSYS writing	0.022 / 0.011	+1.1	+2.10	1.985	1.764	+0.27	-0.025
ShareChat writing	0.066 / 0.034	+3.2	+3.38	2.491	1.757	+3.35	-0.006

2.4 Support forms differentiate constructive engagement

Education theory treats scaffolding as a family of support moves rather than a single uniform intervention. We next examined how different forms of scaffolded support were associated with users' constructive engagement. We decomposed scaffolded support into the six support forms in the annotation scheme (M1–M6) and, among scaffolded conversations, compared conversations containing each form with scaffolded conversations without that form (Fig. 4a). Because support-form labels were non-mutually exclusive, these estimates describe form-level associations rather than effects of isolated instructional moves.

The form-level pattern was sharper than the binary distinction between scaffolded and non-scaffolded support. When stratified by user framing, feedback-like support showed the largest constructive-engagement contrast in both intentional and unintentional conversations, with a stronger association under intentional framing (+14.7 versus +6.7 percentage points; framing difference, +8.0 points; Fig. 4a). Explanation showed the same direction (+6.7 versus +3.0 points; framing difference, +3.6 points), whereas instructing and questioning were lower, especially in intentionally framed conversations. Task-stratified estimates provided a complementary check (Fig. 4b): feedback and explanation were positive in both coding- and writing-oriented ecologies, while instructing, modelling and questioning varied more strongly by task. This pattern suggests that the forms most aligned with constructive participation were those that responded to a user's prior attempt or made mechanisms available for inspection, while more directive, example-based or clarification-oriented moves depended more strongly on the surrounding task ecology. Thus, after the conversation-level association in Section 2.3, the finer drill-down is not that all scaffolding is equally useful, but that support form shapes whether scaffolding leaves a productive role for the user in the work of understanding.

Assistant support-intent labels added little separation because most scaffolded support was cognitively oriented. The more informative distinction was therefore how support was delivered. Figure 4c situates this form-level pattern within task ecology: explanation dominated scaffolded coding turns across the three corpora, whereas writing-oriented support supply was more mixed and more often included instructing, explanation or questioning forms.



c Supply within scaffolded assistant turns (%)

	M1	M2	M3	M4	M5	M6
WC coding	8.8	14.7	29.7	84.0	23.9	6.9
LMSYS coding	4.5	8.2	10.4	79.2	18.3	10.6
SC coding	13.3	12.7	25.7	81.2	18.4	18.6
WC writing	16.6	9.6	59.2	29.5	15.0	6.9
LMSYS writing	11.3	10.9	52.3	26.9	10.3	13.4
SC writing	24.2	23.2	22.9	45.2	23.5	38.1

WC, WildChat; LMSYS, LMSYS Chat; SC, ShareChat. Means labels are non-exclusive.

Fig. 4 Support form and support supply differentiate scaffolded interaction. a, Constructive-engagement differences associated with each support form among scaffolded conversations, stratified by intentional versus unintentional user framing. **b,** The same support-form contrasts stratified by coding- versus writing-oriented task ecology. **c,** Percentage of scaffolded assistant turns containing each support form by task setting. In panels a and b, bars compare scaffolded conversations containing a support form with scaffolded conversations without that form; error bars show 95% confidence intervals, and bracket annotations show between-stratum differences and Benjamini–Hochberg FDR-adjusted q values within each panel family. Support-form labels correspond to M1–M6 and are not mutually exclusive, so rows do not sum to 100%.

Together, these results show why scaffolding presence alone is too coarse to characterize learning-oriented human–LLM interaction. Scaffolded support was associated with richer constructive participation at the conversation level, but that association was carried unevenly by support form and task-specific support supply. The implication is not that systems should simply provide more scaffolding: feedback and explanation may sustain users who are already working with ideas, whereas more directive forms may complete the task without eliciting further elaboration.

2.5 Support–engagement coupling is local and state-dependent

The preceding analyses narrowed the support result: scaffolded conversations were more constructive overall, but the association depended on support form. We finally asked whether this pattern was visible in turn ordering. If scaffolding is part of an interactional process, its clearest signal should appear locally, in how an assistant responds to the user's current state and how the next user turn continues the exchange.

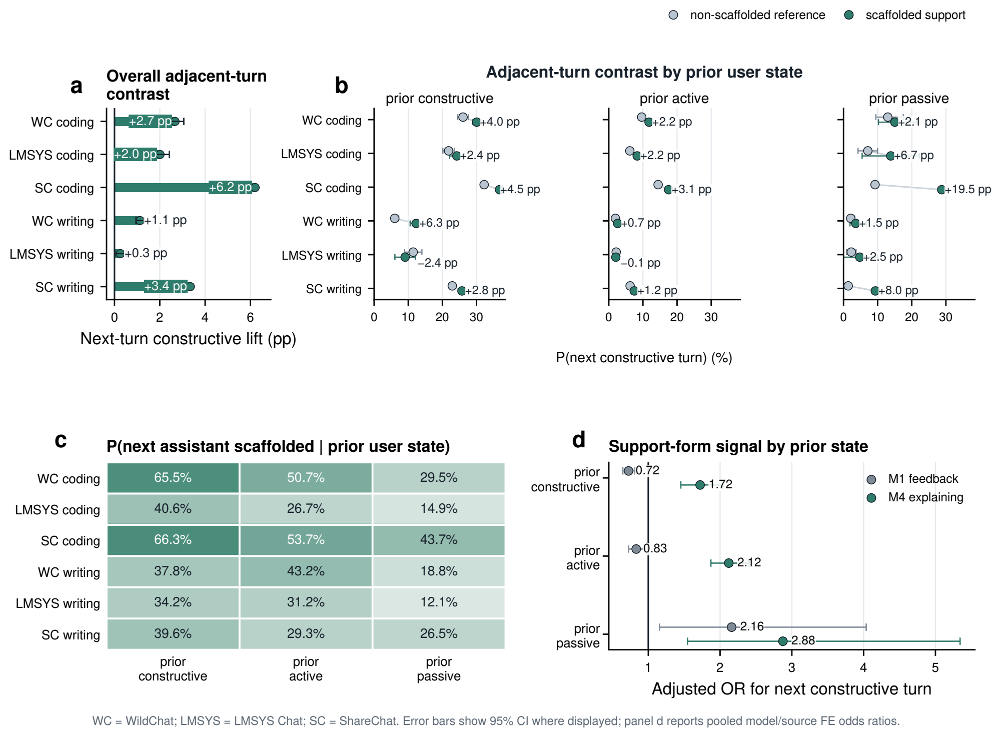


Fig. 5 **Support–engagement coupling is local and state-dependent across the three corpora. a,** Next-turn constructive lift after scaffolded versus non-scaffolded reference assistant turns. **b,** Probability of a constructive next user turn by prior user state and support condition; annotations show scaffolded-minus-reference differences. **c,** Probability that the next assistant turn contains scaffolded support by prior user state. **d,** Pooled model/source fixed-effect odds ratios for focal support forms within prior user states; M1 feedback and M4 explaining are shown because Section 2.4 identified them as the central positive conversation-level forms, and the full M1–M6 interaction table is reported in Supplementary Table C13. Error bars show 95% confidence intervals where displayed. Panel d reports adjacent-turn logistic models with conversation-cluster robust standard errors. WC, WildChat; LMSYS, LMSYS Chat; SC, ShareChat.

At this adjacent-turn scale, scaffolded assistant turns were more likely than non-scaffolded reference turns to be followed by constructive engagement in all six task settings (Fig. 5a). The lift was positive in each setting, ranging from +0.27 to +6.21 percentage points. Five of the six adjacent-turn lifts were statistically distinguishable from zero in conversation-cluster bootstrap tests; LMSYS writing was positive but weaker and did not cross the conventional .05 threshold (+0.27 percentage points, $p =$ .061; Supplementary Table C6). Thus, the temporal evidence supports a local coupling claim: scaffolded support was followed by more constructive next-turn participation, but the magnitude of that coupling varied across settings.

The local pattern was not simply a one-way effect of support on an otherwise stable user. It depended on the user's immediately preceding state. State-conditioned contrasts were generally positive but varied in size across prior constructive, active and passive turns (Fig. 5b). The reverse sequence showed the same contingency: assistant scaffolding was more likely after constructive or active user turns than after passive turns, especially in coding-oriented conversations (Fig. 5c). This bidirectional ordering suggests a coupled process in which user engagement helps elicit scaffolding, and scaffolded support is then associated with further constructive engagement in the next turn.

The support-form analysis also carried into the local model, but with an important refinement. Integrated adjacent-turn regressions showed that scaffolding features still added substantial signal after user state, task/framing, turn position and model/source fixed effects were included (full scaffolding block LR $\chi^2_7 = 1337.5$, $p < .001$; Supplementary Table C12). Adding prior-state × support-form interactions further improved fit (pooled model/source fixed effects: LR $\chi^2_{18} = 300.2$, $p < .001$; Supplementary Table C13). Figure 5d displays M1 feedback and M4 explaining as focal forms because they were the central positive conversation-level forms in Section 2.4; the full M1–M6 interaction results are reported in Supplementary Table C13. In these state-stratified models, explanation was the most stable positive local form: M4 explaining predicted next-turn constructive engagement after prior constructive, active and passive turns (ORs 1.72, 2.12 and 2.88, respectively; all $p < .001$). Feedback was more dependent on the preceding state, consistent with the idea that a support form can have a different immediate role depending on whether it follows an already engaged attempt or a more passive turn.

This local interpretation is also the scale supported by the longer-window checks. Coarser before–after contrasts around the first scaffolded assistant turn are retained in the Appendix only as exploratory boundary checks, not as accumulation estimates, because the first scaffolded turn may itself be selected by task difficulty, uncertainty or the prior trajectory of the exchange (Supplementary Table C5; Supplementary Fig. D1). The main temporal estimand is therefore the one-turn sequence in Fig. 5: prior user state, assistant support form and the immediately following user turn.

Taken together, the temporal analyses turn scaffolding from a simple exposure variable into an interactional process. The strongest evidence is not that support accumulates globally across a conversation, nor that any scaffolded response is equally helpful. It is that constructive engagement emerges when the user's current state and the assistant's support form are locally aligned: engaged users tend to elicit scaffolding, explanatory support is most consistently followed by further constructive work, and other forms shift with the state they respond to. The broader implication is that learning-oriented engagement in everyday human–LLM interaction is not simply delivered by the model, but assembled turn by turn through sequences that preserve reasons, uncertainty and room for the user to keep thinking.

3 Discussion

Everyday human–LLM interaction as an informal learning ecology

This study identifies everyday human–LLM interaction as a middle space between ordinary tool use and designed instruction. Across naturalistic WildChat, LMSYS

Chat and ShareChat coding and writing conversations, learning-oriented engagement was recurrent, but its constructive form was selective. Constructive engagement is useful here because it marks the user-side work that makes informal learning visible in logs: testing a boundary, revising an idea, articulating a constraint or extending an explanation. The Results show that this deeper participation was not simply a property of a user or a model response. It appeared more often when intentional user framing, task affordances, sustained exchange and assistant support preserved reasons and opportunities for users to keep working with ideas. Feedback and explanation aligned more clearly with constructive engagement than more directive or question-based moves, and the temporal evidence was strongest in adjacent-turn coupling. Together, these findings suggest that informal learning in everyday LLM use emerges through situated exchanges in which users and assistants maintain a shared space for testing, revising and extending understanding while pursuing practical goals.

This view differs from treating educational LLMs primarily as systems that imitate designed pedagogy, such as tutoring scripts or Socratic questioning. Recent work on LLM tutors makes a related point from the opposite direction: benchmark settings often assume that students will take up scaffolded guidance, whereas real deployments show that students may redirect or bypass pedagogical framing when it does not fit their goals [23]. Those approaches remain valuable when learning is the explicit goal, but everyday informal learning is often less deliberate. In our results, constructive participation also appeared in task-oriented conversations, increased in sustained exchanges and varied across task ecologies. Coding exchanges, for example, often made uncertainty visible through errors, constraints and executable tests, whereas writing exchanges more often allowed users to apply a revision without articulating why it worked. The implication is not that one task domain is inherently more educational than another. Rather, everyday LLM use creates different opportunities for users to ask, test, refine and extend ideas depending on what the task makes visible and how much responsibility the interaction leaves with the user.

Scaffolding as situated support

The support results show how assistant behaviour becomes part of this ecology. Conversations containing scaffolded support showed richer constructive participation across the six main settings, and adjusted models preserved positive associations after accounting for measured conversation features. These population-level patterns indicate that scaffolded support marks conversational environments in which constructive participation becomes more visible. The key point is not the mere presence of pedagogical language, but how support is positioned within the user's ongoing work, consistent with scaffolding accounts that emphasize contingency, responsiveness and the transfer of responsibility [16, 17, 19]. Scaffolding is most informative when analysed together with what users are trying to do, what the task makes visible and how the assistant response shapes the user's opportunity to continue working with ideas.

The most actionable insight is that support form matters. At the conversation level, feedback-like support and explanation were most clearly aligned with constructive engagement, whereas instructing and questioning were less aligned with visible elaboration. This does not mean that each form has the same local role at every turn: feedback was a strong conversation-level marker but was more state-contingent in adjacent-turn models, whereas explanation was the most stable local support form. This distinction suggests that different support forms leave different roles for the user. Feedback responds to a user's prior attempt and can create an occasion for revision, evaluation or extension. Explanation can make mechanisms visible and give users material for testing their understanding. Instruction and questioning remain important forms of assistance, especially when users need direct task support, missing information or clarification. In the observed conversations, however, these forms were less aligned with visible constructive elaboration, suggesting that support quality for informal learning should be judged by whether it preserves a productive role for the user in the work of understanding, not only by whether it appears pedagogical in form.

From local coupling to lifelong learning support

The temporal results give this scaffolding interpretation a behavioural scale. Scaffolded assistant turns were followed by higher next-turn constructive engagement in all six settings, with five of the six adjacent-turn lifts statistically distinguishable from zero. The reverse ordering also mattered: assistant scaffolding was more likely after constructive or active user turns than after passive turns, especially in coding-oriented conversations. These two directions make the interaction look less like a one-way instructional pipeline and more like a local transfer-of-responsibility process: users' current activity helps elicit support, and some forms of support are then followed by further user work. The prior-state $\times$ support-form models sharpened this point. Explanation was the most stable positive local form, whereas feedback depended more on the preceding user state, suggesting that the same support form can leave different responsibilities to the user depending on when it appears.

This local pattern sets the boundary for lifelong learning claims. The study does not show that everyday LLM support accumulates into durable skill by itself; longer-window checks around the first scaffolded turn were treated as exploratory because support is selected by difficulty, uncertainty and the prior trajectory of the exchange. The stronger claim is human-behavioural: LLMs are becoming part of the ordinary environments in which people encounter uncertainty, test ideas and decide what to try next. The design challenge is therefore not only to make assistants more tutor-like, but to preserve human agency in these everyday sequences. Useful systems should sometimes answer directly, sometimes explain or give feedback, and sometimes step back so that users retain responsibility for evaluation, revision and transfer. The broader promise is a learning ecology in which AI-mediated work supports people's capacity to keep reasoning while solving real problems, rather than merely accelerating task completion.

Scope and future directions

Several boundaries define the scope of these claims. The analyses characterize population-level patterns in three public conversational corpora, WildChat, LMSYS Chat and ShareChat. They make visible where learning-oriented engagement appears, how constructive participation is organized by task ecology and user framing, and how assistant support aligns with engagement across conversation-level, support-form and adjacent-turn timescales. These patterns should be read as observational digital-trace evidence rather than causal estimates [24, 25]. Applying them to individual learners will require additional information about prior knowledge, goals, motivation and interaction history, because the same support form may function differently for different users over time. The labels also capture behavioural manifestations of learning-oriented engagement rather than direct evidence of durable learning outcomes. Future work should connect these behavioural signatures to independent measures of retention, transfer, skill development and learner autonomy, and should test whether support policies that preserve user responsibility improve learning without reducing useful task assistance.

Within these boundaries, the study reframes everyday human–LLM interaction as a middle space between ordinary tool use and formal instruction. Large-scale log analysis can reveal the interactional structure of this space; controlled interventions and longitudinal user-level studies can test how repeated local episodes become durable knowledge, transferable skill or greater learner autonomy. The broader social question is whether AI-mediated work will make people more dependent on fluent answer delivery, or whether it can be shaped into an environment where people continue to build judgment while acting in the world. That question is not only technical. It concerns how future AI systems distribute cognitive responsibility, sustain human agency and make everyday problem solving a site for lifelong learning.

4 Methods

4.1 Study design

This study is an observational analysis of naturalistic public human–LLM conversations from WildChat, LMSYS Chat-1M (reported below as LMSYS Chat) and the ShareChat strict-English subset (reported below as ShareChat) across two everyday task ecologies, coding and writing. It asks whether learning-oriented engagement is visible in ordinary use, where constructive engagement is concentrated and how assistant scaffolding is ordered with user engagement across conversation-level and adjacent-turn timescales. The design is descriptive and associational, following the logic of large-scale digital trace and computational social-science analyses [24, 26]. It does not estimate randomized exposure effects, causal instructional efficacy or durable learning outcomes [25].

The empirical sequence mirrors the Results section. We first estimate the prevalence and composition of user engagement, then examine how constructive engagement varies by user framing, task ecology and interaction depth. We next test whether scaffolded assistant support marks conversations with richer constructive participation, separate scaffolded support by support form and analyse local temporal coupling between assistant support and user engagement. Dataset and task ecology are treated as substantive dimensions of the interaction environment rather than nuisance variables to be averaged away.

4.2 Data sources and analytic samples

The analytic corpus contains 128,334 conversations and 979,974 total turns, including 490,932 user turns and 489,042 assistant turns. The six task settings are WildChat coding, WildChat writing, LMSYS Chat coding, LMSYS Chat writing, ShareChat coding and ShareChat writing. WildChat and LMSYS Chat provide large public human–LLM interaction logs from deployed chat systems, whereas ShareChat provides a smaller publicly shared conversation corpus after English-language filtering; together they offer complementary naturalistic settings rather than experimentally matched platform conditions [11–13]. WildChat coding contributes 31,878 conversations with 124,073 user turns and 124,623 assistant turns. WildChat writing contributes 39,534 conversations with 163,705 user turns and 164,824 assistant turns. LMSYS Chat coding contributes 31,879 conversations with 106,312 user turns and 104,431 assistant turns. LMSYS Chat writing contributes 21,023 conversations with 77,906 user turns and 76,279 assistant turns. ShareChat contributes 2,481 coding conversations with 11,541 user turns and 11,522 assistant turns and 1,539 writing conversations with 7,395 user turns and 7,363 assistant turns.

We also processed two newer corpora with the same pipeline as boundary checks. SWE-chat provides a recent agentic software-engineering setting and retained 1,536 coding conversations after filtering, so we report it as a coding-specific external check rather than as part of the balanced six-setting analysis [27]. ThoughtTrace retained only 56 coding and 61 writing conversations after the final task-and-language filter, with near-saturated scaffolded support, and is therefore kept as a feasibility check in the Appendix [28].

The six task settings were analysed with a common Level 1–4 annotation and analysis framework so that prevalence, contextual variation, support–engagement association and temporal ordering were computed from the same construct definitions. The manuscript reports the refreshed production outputs after the latest user-turn update pass and the subsequent LMSYS Chat and ShareChat annotation runs rather than earlier pre-update batches. Supplementary Section A.1 describes the corpus construction and refresh provenance in more detail.

4.3 Conversation metadata and task ecologies

Each conversation is represented as an ordered sequence of user and assistant turns together with available metadata. The main contextual variables are data source, task domain, user framing, topic and language. Coding and writing were selected because they are common everyday LLM use cases but differ in their interactional affordances: coding often involves debugging, error diagnosis and iterative testing, whereas writing often involves drafting, revision and directive editing [20–22]. User framing distinguishes conversations explicitly oriented toward understanding, exploration or learning from conversations without such explicit learning-oriented framing.

Behavioural depth is measured from the number and ordering of turns in the conversation. Depth is used descriptively, as a marker of interactional opportunity and sustained exchange, not as an independent manipulation. Post-answer depth is measured as the number of turns after the first assistant response.

4.4 Turn-level constructs

The unit of annotation is the conversational turn. User and assistant turns are labelled with different constructs because they play different roles in the interaction. User turns are labelled for behavioural, emotional and cognitive engagement, drawing on engagement theory and the ICAP distinction between passive, active and constructive participation [14, 15]. The main analyses emphasize cognitive engagement and its depth levels. Passive engagement (`P`) captures reception or acknowledgement without substantial transformation. Active engagement (`A`) captures task-relevant action, follow-up or manipulation of supplied information. Constructive engagement (`C`) captures elaboration, revision, testing, extension or transformation of ideas. Constructive engagement is treated as the highest-level behavioural proxy for learning-oriented participation available in these logs, not as a direct measure of knowledge gain.

Assistant turns are first distinguished as `S1` or `S2`. `S1` denotes a non-scaffolded reference response, including straightforward task fulfilment or stand-alone information provision. `S2` denotes scaffolded support: assistant behaviour that structures, regulates or redirects the user's process while leaving some cognitive work with the user. This definition follows the scaffolding tradition, especially contingency, transfer of responsibility and potential fading [16, 17, 19]. Scaffolded turns can also receive intention labels (`I1` metacognitive, `I2` cognitive and `I3` affective support) and non-mutually exclusive means labels (`M1` feedback, `M2` hinting, `M3` instructing, `M4` explaining, `M5` modelling and `M6` questioning). Operational examples are provided in Supplementary Table C1.

4.5 Annotation workflow and verification

The annotation workflow combined theory-grounded codebook development, prompt iteration, parser checks, human review and production-scale LLM-assisted labelling. We used LLM-assisted annotation as a scalable text-analysis workflow while retaining parser checks and human verification because prior work shows both the promise and the need for validation when language models are used as annotators [29, 30]. Prompt and post-processing rules were first developed on validation batches to reduce common failure modes, including over-labelling brief acknowledgements as active engagement and misclassifying directive assistant responses as scaffolding. After the pipeline stabilized, the same annotation logic was applied to the WildChat, LMSYS Chat and ShareChat analytic corpora. All examples reproduced in the manuscript are de-identified and paraphrased rather than copied from raw logs.

Human verification was used as a quality-control input rather than as a separate source of outcome measurement. The final coding validation rerun combines three review rounds, with 658 task turns and 643 turns after deduplication. The final writing validation used 210 reviewed conversations. These validation sets were organized by task ecology rather than independently powered within each public corpus. WildChat also underwent a final targeted user-turn update check; LMSYS Chat and ShareChat were then processed with the same codebook, parser checks and post-processing logic, but the current artifacts do not include separate human-agreement estimates for each added corpus. Agreement was summarized with per-label F1, Matthews correlation coefficient and Gwet's AC1 because the labels are sparse, prevalence-imbalanced and not mutually exclusive in all cases [31, 32]. Agreement was strongest for constructive, passive, emotional and support-form labels; boundary-heavy labels such as writing `A` and `S1` were less stable. Supplementary Table C2 reports the per-label agreement metrics, and Supplementary Section A.4 describes the final user-turn update check and corpus-transfer scope.

4.6 Outcome construction

The primary user-side outcome is constructive engagement. At the turn level, this is a binary indicator for whether the user turn is labelled `C`. At the conversation level, we use both the count of constructive user turns and an indicator for whether a conversation contains at least one constructive user turn. Constructive ratios are defined as constructive user turns divided by user turns.

The primary assistant-side exposure is scaffolded support. At the turn level, this is the distinction between `S2` scaffolded support and `S1` non-scaffolded reference response. At the conversation level, we use an indicator for whether a conversation contains at least one `S2` assistant turn. Support-form analyses condition on scaffolded turns and compare conversations containing each means label with scaffolded conversations that do not contain that label. Because means labels can co-occur, these are descriptive support-form contrasts rather than mutually exclusive treatment comparisons.

4.7 Statistical analyses

Descriptive analyses estimate engagement prevalence, scaffolding prevalence, passive/active/constructive composition, emotional-engagement rates and behavioural-depth summaries within each task setting. Context analyses compare intentional versus unintentional conversations and coding versus writing settings. Depth analyses group conversations into behavioural-depth buckets and estimate the probability that a conversation contains at least one constructive turn. Because any-event outcomes mechanically become easier to observe in longer conversations, we also fit a grouped-binomial constructive-rate sensitivity model in which the outcome is the constructive user-turn count out of total user turns, with user framing, task ecology, length bucket and dataset fixed effects as predictors.

Conversation-level associations are analysed by comparing conversations with and without at least one scaffolded assistant turn. Constructive-ratio contrasts use turn-weighted ratios within each comparison group, defined as total constructive user turns divided by total user turns, with uncertainty from 1,000 conversation-level bootstrap resamples. The adjusted Section 2.3 count model is a setting-specific Poisson generalized linear model for the raw constructive-turn count per conversation (`Cog_C_count`), not for a rate. We did not use an offset such as `log(user_turns)` in the primary model; instead, conversation length entered the model directly as `total_turns`. The full Poisson specification was `Cog_C_count` $\sim$ `has_S2` + `is_intentional` + `is_coding_topic` + `total_turns` + `Emo_ratio` + `has_error` + `persistence_after_failure` + `high_persistence`, with constant or non-estimable terms omitted within settings when necessary. The coefficient for `has_S2` is reported after exponentiation as a multiplicative association with expected constructive-turn count conditional on observed covariates. As a rate sensitivity, we refitted the same setting-specific mean model with `offset(log(user_turns))` and quasi-Poisson scaled standard errors.

The companion binary model for Section 2.3 is a setting-specific logistic regression for whether a conversation contains at least one constructive user turn (`has_Cog_C`), using `has_S2`, `is_intentional`, `is_coding_topic` and `total_turns`. Model-based standard errors and 95% confidence intervals are shown for these adjusted Poisson and logistic estimates. We checked Poisson overdispersion using the Pearson χ^2 statistic divided by residual degrees of freedom; the dispersion estimates ranged from 1.16 to 2.06 across the six settings. A quasi-Poisson sensitivity that scaled the Poisson standard errors by this dispersion preserved the direction and statistical distinguishability of the `has_S2` association in every setting, and the offset-rate sensitivity likewise preserved positive `has_S2` associations in every setting. Negative-binomial models were not used as the primary specification because the estimand was the common pre-specified multiplicative association under the same mean model across settings; the overdispersion and offset checks were used to assess variance and exposure-specification sensitivity rather than to change the main estimand. Binary outcomes in other analyses, including whether the next user turn is constructive and whether the next assistant turn is scaffolded, are modelled with logistic regression or reported as conditional probability contrasts, depending on the estimand [33, 34]. Percentage-point contrasts are reported as effect sizes; adjacent-turn contrasts use 1,000 conversation-cluster bootstrap resamples because multiple turn pairs can come from the same conversation. Support-form bracket tests in Fig. 4 use Benjamini–Hochberg false-discovery-rate adjustment within each displayed family of six support-form comparisons. Supplementary Tables C3–C5 summarize the main model families and temporal estimands.

To assess whether the adjacent-turn pattern was reducible to a single feature family, we also fitted integrated logistic regressions that combined user-state features, assistant-scaffolding features and adjacent-turn context in one model. The outcome was whether the next user turn was constructive. The pooled model included dataset fixed effects; a sensitivity replaced these with recoverable assistant model or source-family fixed effects, using exact `chat_model` labels where available, LMSYS Chat model names recovered from conversation identifiers and ShareChat assistant/source families recovered from conversation identifiers. Standard errors for these adjacent-turn regressions were clustered by conversation. We report broad S2 models separately from support-form decompositions because M1–M6 are nested descriptors of scaffolded support rather than independent exposures. Nested likelihood-ratio block tests first add broad scaffolded-support presence and then test whether support-form descriptors add signal within scaffolded support. These model/source checks are interpreted as robustness analyses rather than model-causal comparisons.

Temporal analyses focus on adjacent-turn ordering. The primary analyses compare the probability that the next user turn is constructive after scaffolded versus non-scaffolded assistant turns, estimate the same contrast within prior user states and estimate the probability that the next assistant turn is scaffolded after different prior user states. We also fit adjacent-turn regressions with prior-state $\times$ support-form interactions to test whether local coupling depends jointly on the user's immediately preceding engagement state and the form of scaffolding provided. Coarser within-conversation before–after summaries are reported only as exploratory boundary checks, not as evidence of cumulative learning trajectories, because the first scaffolded turn may be selected by task difficulty, user motivation or an already developing exchange.

4.8 Robustness, privacy and LLM use

WildChat robustness analyses repeat the main patterns across user-facing model snapshots and coarse model families. These checks assess whether the headline patterns are reducible to a single WildChat model label. They are not interpreted as model-causal comparisons, because model labels are not experimentally assigned and may be confounded with time, user population, task mix and logging context.

The analyses are bounded by the observational design and by the nature of the labels. They support claims that everyday human–LLM conversations contain measurable learning-oriented engagement; that constructive engagement varies by intent, task ecology and depth; that assistant scaffolding is associated with richer constructive participation; and that support and engagement show local reciprocal ordering. They do not support claims that scaffolded support causes durable learning gains, that all constructive engagement reflects actual knowledge change or that the labelled conversational states perfectly measure learning itself.

LLMs were used as objects of study and as annotation tools. Production labels were generated with LLM-assisted workflows and then checked through parser rules, targeted update scripts and human validation. The authors also used LLM-based tools to assist with code drafting, debugging, analysis workflow organization and language editing. LLM-generated assistance was not used as a substitute for human verification of the validation samples or for interpretation of the statistical results.

All reported results are aggregate turn-level or conversation-level analyses. The study does not link conversations into user histories, estimate user-level longitudinal trajectories or publish raw conversation text; temporal analyses are confined to ordering within a conversation.

Data availability

This study analyses public human–LLM conversation corpora released by their original data providers: WildChat, LMSYS Chat-1M, ShareChat, SWE-chat and ThoughtTrace. Readers should obtain the raw public conversation data from the original releases and comply with the corresponding dataset licences, terms of use and redistribution restrictions. The manuscript does not redistribute raw message text, user identifiers or linked user histories.

Derived data sufficient to verify the reported aggregate figures, tables and statistical summaries are provided with the analysis repository as figure source-data files in `source_data/`, table files in `tables/` and model/statistical output files in `outputs/integrated_regression/`. These files include the numeric values underlying the main figures, table summaries, confidence intervals, p values, false-discovery-rate q values and model-output summaries used in the manuscript. Before submission or publication, these derived data will be deposited in a DOI-minting repository such as Zenodo or OSF, and the accession link will be inserted here. Any request for additional derived label files that are not included in the public deposit will be assessed against the original dataset licences and privacy constraints.

Code availability

Custom code used to filter corpora, run the LLM-assisted annotation workflow, post-process labels, compute statistical analyses and generate figures and tables is maintained in the project repository at https://github.com/CinderD/Jun_10_NHB_Informal_Learning. A frozen archival release with a DOI will be created before submission or publication and cited here. The repository excludes API keys, raw message text and any files that cannot be redistributed under the original corpus terms. Reproducing the full annotation pipeline requires user-provided API credentials and access to the raw public corpora from their original sources.

Ethics statement

The study is a secondary analysis of public, naturalistic human–LLM conversation datasets. The authors did not recruit participants, intervene in conversations or attempt to identify users. Analyses are reported at aggregate turn or conversation level, and examples in the manuscript are paraphrased rather than copied from raw logs. Human validation was used to assess annotation quality on reviewed examples and did not involve interaction with the original dataset users. Institutional ethics or IRB determination should be confirmed before submission; approval or exemption details will be inserted here if required by the submitting institution.

Competing interests

The authors declare no competing interests.

Author contributions

Author contributions will be finalized after the author list is confirmed. The expected contribution categories are: conceptualization, study design, data curation, annotation pipeline development, validation, statistical analysis, figure generation, writing of the original draft, manuscript review and editing, supervision and funding acquisition. All listed authors will approve the submitted version and will be accountable for their contributions.

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LLM use statement

Large language models were objects of study and were also used as annotation and writing-assistance tools. Production labels were generated with LLM-assisted workflows and checked through parser rules, targeted update scripts and human validation. The authors also used LLM-based tools to assist with code drafting, debugging, workflow organization and language editing. No LLM or AI tool is listed as an author, and the authors take responsibility for the final manuscript, analyses and interpretation. Additional details are provided in Methods.

Reporting summary and editorial policy checklist

A completed Nature Portfolio Reporting Summary and any journal-required editorial policy checklist will be provided at submission. The manuscript and supplementary information report the observational design, sampling and filtering rules, annotation workflow, human validation evidence, statistical model specifications, bootstrap procedures, multiple-comparison handling and data/code availability statements needed to assess reproducibility.

Figure source data

Source data for the numeric main figures are provided in `source_data/figure2_source_data.csv`, `source_data/figure3_source_data.csv`, `source_data/figure4_source_data.csv` and `source_data/figure5_source_data.csv`. Figure 1 is a conceptual framework figure and has no numeric source data. Supplementary table and regression source files are provided in `tables/` and `outputs/integrated_regression/`. These files will be included in the public data/code deposit before submission or publication.

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Supplementary information

Supplementary information is provided below.

Appendix A Supporting methods

A.1 Corpus construction and analytic settings

The analyses use six task settings formed by crossing three public conversational corpora with two task ecologies: WildChat coding, WildChat writing, LMSYS coding, LMSYS writing, ShareChat coding and ShareChat writing. Coding and writing were selected because both are common everyday LLM use cases, yet they differ in the opportunities they create for visible learning-oriented engagement. Coding conversations often contain error messages, debugging attempts, execution failures and iterative repair. Writing conversations more often involve drafting, rewriting, shortening, tone adjustment and direct editing.

The analytic corpus contains 128,334 conversations and 979,974 total turns. WildChat contributes 31,878 coding conversations and 39,534 writing conversations; LMSYS contributes 31,879 coding conversations and 21,023 writing conversations; ShareChat contributes 2,481 coding conversations and 1,539 writing conversations. All reported analyses use the refreshed Level 1–4 outputs after the latest user-turn update pass and the subsequent LMSYS and ShareChat annotation runs.

SWE-chat and ThoughtTrace were also processed with the same pipeline as feasibility and boundary checks (Supplementary Table C15) [27, 28]. SWE-chat retained a usable coding sample (1,536 conversations) but only three writing conversations after filtering, so it can serve only as a coding-specific stress test rather than as part of the main coding–writing comparison. ThoughtTrace retained 56 coding and 61 writing conversations after the final task-and-language filter, but the sample remains small and scaffolded support is near-saturated, especially in coding where all conversations contain at least one scaffolded assistant turn. These properties make ThoughtTrace useful as a feasibility check but not as stable evidence for conversation-level support-association estimation.

For each conversation, the pipeline retained the ordered turn sequence, role labels, available topic and language metadata, learning-intent indicators and model metadata where available. Model metadata are used only for robustness checks where the corpus exposes usable model labels, because model assignment in naturalistic public conversations is observational and may reflect time, user population, task mix and logging context.

A.2 Annotation ontology

The codebook translates learning-science constructs into role-specific behavioural signatures. User turns are labelled for engagement. Passive engagement (`P`) marks acknowledgement, reception or minimal continuation without visible transformation. Active engagement (`A`) marks task-relevant action or follow-up, such as asking for a reformatted answer or applying supplied information. Constructive engagement (`C`) marks deeper work with ideas, such as testing an implication, revising a hypothesis, articulating a constraint, transferring a mechanism or extending an explanation. Emotional engagement is labelled separately because affective expressions were rare and not the main analytic outcome.

Assistant turns are labelled for support. `S1` denotes a non-scaffolded reference response: the assistant provides the requested information or deliverable without materially structuring the user's process. `S2` denotes scaffolded support: the assistant structures, regulates, diagnoses or redirects the user's work while leaving some responsibility with the user. Scaffolded turns can also receive support-intent labels and support-means labels. Intention labels distinguish metacognitive support (`I1`), cognitive support (`I2`) and affective support (`I3`). Means labels distinguish feedback (`M1`), hinting (`M2`), instructing (`M3`), explaining (`M4`), modelling (`M5`) and questioning (`M6`). Means labels are non-mutually exclusive because a single assistant turn can both explain a mechanism and provide an example or hint. Supplementary Table C14 reports the support-means supply profile across the six task settings.

A.3 LLM-assisted annotation pipeline

The annotation pipeline combined prompt-based labelling with deterministic parser checks and targeted post-processing. During development, prompts were iterated against validation examples and disagreement cases. Common failure modes included treating brief acknowledgements as active engagement, treating any long assistant answer as scaffolding, collapsing directive instructions into scaffolding without checking whether responsibility remained with the user, and over-labelling sparse support means.

Production labels were generated at turn level. Parser checks enforced the expected label schema, normalized label variants and flagged missing or malformed outputs. Post-processing rules were used only when a recurring error pattern could be specified transparently, such as a narrow class of user-turn false positives. Prompt versions, run metadata, disagreement files and update outputs were retained with the analysis artifacts so that validation cases can be traced without reproducing raw conversation text in the manuscript.

A.4 Human verification and update provenance

Human verification was used to quantify label reliability and identify boundary cases. Coding validation combined three review rounds, with 658 task turns and 643 turns after deduplication. Writing validation used 210 reviewed conversations. Because many labels are sparse or prevalence-imbalanced, agreement is reported with per-label F1, Matthews correlation coefficient and Gwet's AC1 rather than with a single global agreement statistic.

The final user-turn check contained 100 coding examples and 100 writing examples reviewed after the update pass. These examples were used to identify residual false positives and calibrate patch-style update filters, especially for Active and Passive user-turn labels. The last WildChat user-turn update pass scanned 31,878 coding conversations and 39,534 writing conversations, applying 1,815 coding updates and 21 writing updates. LMSYS Chat and ShareChat were then processed with the same refreshed annotation workflow, construct definitions and post-processing logic. The main manuscript reports the refreshed labels produced by these final production runs.

The validation evidence is therefore domain-stratified rather than fully corpus-stratified. It supports the reliability of the construct definitions and the final coding/writing labelling logic, but it should not be read as separate human-agreement evidence for WildChat, LMSYS Chat and ShareChat individually. Cross-corpus use of the labels is supported by applying the same schema, parser checks and post-processing rules to all three corpora and by reporting setting-level robustness checks in the Results and Supplementary Tables, rather than by a separate manual audit for each added corpus.

A.5 Outcome construction

At the turn level, the focal user-side outcome is a binary indicator for constructive engagement. At the conversation level, we use both the count of constructive user turns and an indicator for whether at least one constructive user turn occurs. Constructive ratios divide constructive user turns by total user turns; group-level constructive-ratio contrasts are turn-weighted ratios computed from total constructive user turns over total user turns within each group. Cognitive engagement ratios include passive, active and constructive cognitive engagement.

At the assistant side, scaffolded support is represented by the `S2` label. Conversation-level scaffolding presence indicates whether at least one assistant turn in the conversation is labelled `S2`. Support-form analyses condition on conversations or turns with scaffolded support and then compare the presence versus absence of each means label. These comparisons are descriptive because means labels can co-occur and because conversations are not randomly assigned to receive a support form.

Behavioural depth is computed from turn counts and turn ordering. Post-answer depth counts turns after the first assistant response. Persistence after failure is analysed only for coding-oriented conversations because debugging failures, error messages and failed-execution contexts are more directly observable in coding than in writing.

A.6 Statistical model specification

The main analyses use simple estimands matched to the claim being made. Descriptive prevalence analyses report proportions within the six task settings. Context analyses compare user framing, task ecology and depth groups. Conversation-level support associations use Poisson generalized linear models for constructive-turn counts and logistic models for binary constructive-engagement outcomes. Adjacent-turn analyses use assistant-to-user and user-to-assistant turn pairs to estimate local conditional probabilities.

For the Section 2.3 adjusted conversation-level models, the Poisson outcome is the raw count of constructive user turns in a conversation (`Cog_C_count`), not a rate. The primary model does not include a `log(user_turns)` offset. Conversation length is adjusted directly through `total_turns`, so the exponentiated Poisson coefficient should be read as a multiplicative association with expected constructive-turn count conditional on observed conversation length and other covariates. The full Poisson specification is `Cog_C_count` $\sim$ `has_S2` + `is_intentional` + `is_coding_topic` + `total_turns` + `Emo_ratio` + `has_error` + `persistence_after_failure` + `high_persistence`; constant or non-estimable terms are omitted within settings when necessary. The companion logistic specification is `has_Cog_C` $\sim$ `has_S2` + `is_intentional` + `is_coding_topic` + `total_turns`. The user-framing stratified Poisson models use the same Poisson specification after stratifying by framing and omitting `is_intentional`.

Covariates are included when they correspond to observed compositional differences at the analysis level. Depending on the analysis, these include user framing, topic or task indicators, turn counts, emotional ratios, error-related variables and turn index. Percentage-point contrasts are reported as effect sizes. Confidence intervals and p values for these contrasts use 1,000 conversation-level bootstrap resamples; adjacent-turn contrasts use 1,000 conversation-cluster bootstrap resamples rather than individual turn-pair resampling to account for repeated pairs within conversations. The adjusted Poisson and logistic estimates in Fig. 3 use model-based standard errors. Poisson overdispersion was checked with the Pearson χ^2 statistic divided by residual degrees of freedom. Dispersion ranged from 1.16 to 2.06 across the six task settings; scaling the Poisson standard errors with a quasi-Poisson dispersion adjustment preserved positive `has_S2` associations and statistical distinguishability in every setting. To check exposure specification, we also refitted the same setting-specific Poisson mean model with `offset(log(user_turns))`; Supplementary Table C7 reports quasi-Poisson scaled rate ratios. Negative-binomial models were not used as the primary specification; the sensitivity analyses focus on variance misspecification and exposure specification under the same mean model rather than changing the main estimand. Support-form bracket tests use Benjamini–Hochberg false-discovery-rate adjustment within each six-form comparison family.

The Section 2.2 constructive-rate sensitivity uses a grouped-binomial logistic model at the conversation level. The numerator is the number of constructive user turns and the denominator is total user turns in the conversation. The predictors match the any-constructive context model: user framing, task ecology, length bucket and dataset fixed effects. Standard errors use a conversation-robust sandwich estimator. This sensitivity is included because any-constructive-turn outcomes partly reflect the number of turns available for observing at least one constructive turn.

The integrated adjacent-turn regression combines three feature groups in one logistic model: user state before the assistant turn, assistant scaffolding features, and adjacent-turn context. The pooled specification includes dataset fixed effects. A model/source sensitivity uses exact WildChat model labels, LMSYS model names recovered from conversation identifiers and ShareChat assistant/source families recovered from conversation identifiers. These adjusted estimates are interpreted as measured-covariate-adjusted associations. They do not remove unobserved user motivation, prior expertise, task difficulty or time-varying conversational state.

A.7 Temporal scale

The temporal analyses focus on adjacent-turn ordering because it is the scale at which the logs most directly identify sequence. The adjacent-turn analyses ask whether the immediately following user turn is more likely to be constructive after scaffolded support than after a non-scaffolded reference response, whether that contrast depends on the prior user state and whether support-form signals differ across prior user states. The reverse adjacent-turn analysis asks whether the assistant is more likely to provide scaffolded support after constructive, active or passive user turns.

Coarser before–after summaries ask a different question: whether constructive engagement is higher after the first scaffolded assistant turn than before it. These are retained only as exploratory boundary checks, not as treatment-effect or accumulation estimates. The first scaffolded turn may occur precisely because the user is struggling, because a difficult task has emerged or because the exchange is already developing. The main temporal claim is therefore scale-specific: the clearest evidence concerns local, state- and form-conditioned coupling rather than a demonstrated global trajectory.

Appendix B Robustness and interpretive boundaries

B.1 WildChat model-snapshot robustness

WildChat includes user-facing model metadata that make it possible to repeat major descriptive patterns across model snapshots and coarse model families. These analyses preserved the main directional patterns reported in the main text: coding generally showed higher cognitive engagement than writing; intentionality gaps were visible across most user-facing models; scaffolded conversations were generally more constructive; and adjacent-turn contrasts were more interpretable than coarser within-conversation contrasts. Supplementary Figures D3–D7 report these WildChat model-snapshot checks.

These checks are used as robustness analyses rather than model-causal comparisons. WildChat model labels are not experimentally assigned. They may be confounded with calendar time, user population, task mix, interface changes and logging context. For this reason, the manuscript does not claim that a specific model family caused the observed engagement differences.

B.2 Cross-dataset and model/source regression checks

The Results claims were checked for cross-dataset consistency at the level appropriate to each estimand. Section 2.1 is a descriptive prevalence claim: cognitive engagement, constructive engagement and scaffolded support were observable in all six settings, with constructive engagement consistently the highest-level and less frequent user-side signal. Section 2.2 is primarily descriptive: user framing, task ecology and interaction depth organize where constructive engagement appears, with the same directional depth gradient across settings. As a simple model-based check, we fitted a conversation-level logistic regression for whether a conversation contained at least one constructive user turn, using user framing, task ecology, length bucket and dataset fixed effects (Supplementary Table C8). Because this any-constructive outcome is partly affected by the number of user turns available for observing at least one constructive turn, we also fitted a grouped-binomial constructive-rate sensitivity model, using constructive user-turn counts out of total user turns as the outcome and the same contextual predictors (Supplementary Table C9). Formal uncertainty estimates are otherwise reported where the claim involves a contrast or model coefficient. Section 2.3 uses conversation-level bootstrap confidence intervals and p values for turn-weighted scaffolded versus non-scaffolded constructive-ratio contrasts, adjusted Poisson/logistic model estimates and an offset-rate Poisson sensitivity using user turns as the exposure (Supplementary Table C7). Section 2.4 reports support-form confidence intervals and between-stratum FDR-adjusted q values in the main figure. Section 2.5 reports conversation-cluster bootstrap confidence intervals for adjacent-turn lifts and cluster-robust regression estimates for the integrated local models.

The main percentage-point contrasts were directionally consistent across the six settings. Constructive-ratio contrasts comparing conversations with versus without scaffolded support were positive and statistically distinguishable from zero in all six settings, with effect sizes ranging from +0.99 to +7.34 percentage points. Adjacent-turn constructive lifts were also positive in all six settings, with five of six contrasts statistically distinguishable from zero; LMSYS writing was positive but weaker (+0.27 percentage points, 95% CI -0.01 to +0.56, $p = .061$). Supplementary Table C6 reports these effect sizes, confidence intervals and p values.

Integrated adjacent-turn regressions were used as robustness checks rather than as the primary narrative claim. These models predicted whether the next user turn was constructive after combining user-state features, assistant-scaffolding features and adjacent-turn context. Because model metadata differ by corpus, we report both within-setting models and pooled specifications. The within-setting models show the dataset-by-task pattern directly, with recoverable model/source fixed effects added separately inside each setting (Supplementary Table C10). The pooled specification includes dataset fixed effects; a sensitivity replaces these with exact WildChat model labels, LMSYS model names recovered from conversation identifiers and ShareChat assistant/source families recovered from conversation identifiers. This is interpreted as model/source adjustment rather than as a causal comparison among model versions. Model/source heterogeneity was detectable, but it did not absorb the scaffolding signal. Adding broad S2 support to user/context/model controls improved fit (LR $\chi^2_1 = 509.5$, $p < .001$). Because M1–M6 are support forms nested within S2 rather than independent covariates, we interpret the decomposed models as asking which forms carry additional signal among scaffolded turns, not as a test of S2 after removing its own components. Adding M1–M6 descriptors within scaffolded support improved fit (LR $\chi^2_6 = 827.9$, $p < .001$), and the full scaffolding block remained strongly informative after controls (LR $\chi^2_7 = 1337.5$, $p < .001$; Supplementary Table C12). Adding prior-state $\times$ support-form interactions also improved fit (pooled model/source fixed effects: LR $\chi^2_{18} = 300.2$, $p < .001$; Supplementary Table C13). Fully decomposed coefficients clarify the source of this form-level signal: prior user state is the largest predictor of immediate constructive follow-up, and M4 explaining is the most stable positive support-form coefficient (Supplementary Table C11).

B.3 Label uncertainty

The label scheme intentionally separates top-level constructs from finer support forms. Top-level labels such as constructive engagement and scaffolded support have clearer behavioural anchors. Fine-grained labels, especially sparse or boundary-heavy labels, are less stable. This is why the main claims rely most heavily on constructive engagement, scaffolded support presence, support-form patterns that are consistent across settings and temporal contrasts that can be expressed with simple conditional probabilities.

B.4 Selection into support

The most important threat to causal interpretation is selection into support. More complex tasks, more motivated users or already-developing exchanges may both elicit scaffolded support and produce constructive follow-up. Conversation-level adjusted models reduce measured compositional differences but cannot remove unobserved selection. Adjacent-turn analyses reduce the temporal window and condition on local state, but they still cannot identify randomized exposure effects. The safest interpretation is therefore local, state-dependent coupling between assistant support and user engagement.

B.5 Privacy-preserving reporting

The study analyses naturalistic conversations, so the manuscript avoids reproducing raw conversation text. Operational examples are de-identified and paraphrased to illustrate decision rules without exposing user content. Aggregate statistics, model summaries and validation metrics are reported in the manuscript and supplementary tables. The analyses do not link conversations into user histories or estimate longitudinal user-level trajectories; temporal claims are restricted to within-conversation turn ordering.

Appendix C Supporting tables

Table C1 Operational examples for turn-level labels. Examples are de-identified and paraphrased from validation review patterns; they illustrate the decision rule rather than reproducing raw conversation text.

Construct	Paraphrased turn	Annotation rationale
`P` passive user engagement	"Thanks, that makes sense."	Minimal uptake or acknowledgement without new reasoning, evaluation or task transformation.
`A` active user engagement	"Can you convert that answer into a command I can run in my setup?"	Task-relevant follow-up or manipulation of supplied information, but without elaborating or testing an idea.
`C` constructive user engagement	"If the cache expires before the asynchronous callback returns, would the same race condition still happen?"	The user tests an implication, boundary condition or transfer of the prior explanation.
`S1` non-scaffolded reference	"Here is the requested function/email draft."	The assistant primarily completes the requested deliverable without regulating the user's process.
`S2` scaffolded support	"First isolate the failing step; that will show whether the issue is parsing or state management."	The assistant structures the user's process and leaves diagnostic or revision work with the user.
`M1` feedback	"Your diagnosis is close; the bug is in the index offset, not the loop boundary."	Evaluative feedback on the user's prior attempt or understanding.
`M2` hinting	"Check the parser log around the first malformed token before changing the model."	A cue or partial path that helps the user proceed without fully resolving the task.
`M3` instructing	"First split the condition into two branches, then add a regression test."	Explicit steps or strategy that organizes the user's work.
`M4` explaining	"The exception occurs because the event loop is already running, so a nested call cannot acquire it."	A rationale or mechanism that clarifies why the issue occurs.
`M5` modeling	"A small pattern you can adapt is: validate input, normalize it, then pass it to the writer."	A sample approach intended to be emulated or adapted rather than merely copied.
`M6` questioning	"Which constraint matters more here: preserving tone or shortening the paragraph?"	A question that advances problem solving or reflection.

Table C2 Domain-stratified human–human agreement for labels used in the main analyses. Coding uses the latest three-round validation rerun over 658 task turns, 643 after deduplication; writing uses the latest writing-210 best-pipeline validation. These validation sets assess the coding and writing labelling logic and are not separately powered within each public corpus. F1, Matthews correlation coefficient (MCC) and Gwet's AC1 are reported per binary label. Coding MCC and AC1 are recomputed from the exported support and positive-count fields; writing metrics are from the exported agreement report. Rare degenerate labels with no positive cases in a validation set are omitted.

Domain	Role	Label	F1	MCC	Gwet AC1	Support
Coding	User	Active (`A`)	0.75	0.72	0.87	315
Coding	User	Constructive (`C`)	0.85	0.82	0.90	315
Coding	User	Passive (`P`)	0.88	0.87	0.96	315
Coding	User	Emotional (`E`)	0.83	0.82	0.97	315
Coding	Assistant	Non-scaffolded reference (`S1`)	0.78	0.73	0.88	328
Coding	Assistant	Scaffolded support (`S2`)	0.87	0.69	0.70	328
Coding	Assistant	Feedback (`M1`)	0.90	0.89	0.95	328
Coding	Assistant	Hinting (`M2`)	0.79	0.75	0.89	328
Coding	Assistant	Instructing (`M3`)	0.81	0.77	0.92	328
Coding	Assistant	Explaining (`M4`)	0.86	0.79	0.84	328
Coding	Assistant	Modeling (`M5`)	0.84	0.79	0.87	328
Coding	Assistant	Questioning (`M6`)	0.82	0.81	0.95	328
Writing	User	Active (`A`)	0.38	0.38	0.69	90
Writing	User	Constructive (`C`)	0.89	0.83	0.86	90
Writing	User	Passive (`P`)	0.80	0.81	0.99	90
Writing	User	Emotional (`E`)	0.67	0.65	0.96	90
Writing	Assistant	Non-scaffolded reference (`S1`)	0.46	0.43	0.86	120
Writing	Assistant	Scaffolded support (`S2`)	0.97	0.80	0.93	120
Writing	Assistant	Feedback (`M1`)	0.89	0.80	0.80	120
Writing	Assistant	Hinting (`M2`)	0.82	0.70	0.71	120
Writing	Assistant	Instructing (`M3`)	0.71	0.67	0.86	120
Writing	Assistant	Explaining (`M4`)	0.84	0.70	0.70	120
Writing	Assistant	Modeling (`M5`)	0.94	0.91	0.95	120
Writing	Assistant	Questioning (`M6`)	0.96	0.96	0.99	120

Table C3 Analytic model families used in the main analyses. The table summarizes descriptive and conversation-level model families. Adjusted models are interpreted as covariate-adjusted associations in naturalistic data, not as causal treatment-effect estimates.

Analysis family	Unit and outcome	Main contrast or model	Scope
Engagement prevalence	User turns and conversations; cognitive, constructive and emotional engagement indicators.	Descriptive proportions by task setting, task ecology and user framing.	Establishes where learning-oriented engagement is visible; no causal contrast is implied.
Engagement composition	Cognitively engaged user turns; passive, active and constructive depth levels.	Composition of cognitive engagement within each task setting.	Motivates constructive engagement as the focal high-level outcome while preserving the broader cognitive-engagement hierarchy.
Interaction depth	Conversations; at least one constructive user turn.	Probability of constructive engagement across behavioural-depth buckets.	Depth is interpreted as interactional opportunity and task development, not as a manipulated exposure.
Scaffolding presence	Conversations; constructive-turn count and any constructive turn.	Poisson GLM for `Cog_C_count` with `has_S2`, user framing, task/topic indicator where estimable, total turns, emotional ratio, error markers and persistence markers; no log-turn offset. Logistic model for `has_Cog_C` with `has_S2`, user framing, task/topic indicator and total turns.	Adjusted estimates reduce measured compositional differences but remain associational because support is selected into by task difficulty and user state. Pearson overdispersion was checked and quasi-Poisson scaling preserved the direction of the Poisson estimates.

Table C4 Additional analytic model families and robustness checks. These analyses qualify the main associations by support form, temporal scale and available model or source metadata.

Analysis family	Unit and outcome	Main contrast or model	Scope
Support-form associations	Scaffolded conversations; constructive engagement ratios.	Mean differences for conversations containing a given support-means label versus scaffolded conversations without that label, summarized by user framing and task ecology.	Descriptive because means labels are non-exclusive and can co-occur in the same assistant turn.
Local temporal coupling	Adjacent assistant–user or user–assistant turn pairs; next constructive user turn or next scaffolded assistant turn.	Conditional probability contrasts around adjacent turns; detailed estimands are listed in Supplementary Table C5.	Interpreted as local ordering and state-dependent coupling, not as randomized support effects.
Integrated adjacent-turn regression	Adjacent assistant–user turn pairs; next constructive user turn.	Logistic regression combining prior user state, scaffolded-support presence, support means, task/framing context, prior-state $\times$ support-form interactions and dataset or model/source fixed effects.	Tests whether the local coupling pattern depends jointly on user state and support form; standard errors are clustered by conversation.
Coarse temporal boundary checks	Conversations containing scaffolded support; constructive engagement before and after the first scaffolded turn.	Mean after-minus-before contrast reported as an exploratory appendix check.	Not used as evidence of accumulation because the first scaffolded turn may be selected by difficulty, uncertainty or an already developing exchange.
WildChat model robustness	WildChat conversations or turns; main prevalence, association and temporal outcomes.	Repetition across user-facing model snapshots or coarse model families.	Robustness summaries only; model labels are observational and may be confounded with time, user mix and task mix.

Table C5 Temporal estimands used to characterize local support–engagement coupling. The main temporal analyses use adjacent assistant–user or user–assistant turn pairs. Coarser before–after summaries are retained only as exploratory boundary checks because scaffolded turns are selected by task difficulty, user motivation and conversation trajectory.

Estimand	Definition	Reported in main text	Scope
Overall adjacent-turn lift	Difference in the probability that the next user turn is constructive after scaffolded support versus after a non-scaffolded reference response, estimated within each task setting.	Figure 5a	Local association between assistant support type and immediately following constructive engagement; not a randomized support effect.
Prior-state adjacent contrast	Difference in next-turn constructive probability after scaffolded support versus reference response, separately for prior constructive, active and passive user states.	Figure 5b	Tests whether local support–engagement coupling depends on the user's immediately preceding state.
Reverse state-conditioned support	Probability that the next assistant turn is scaffolded, estimated after constructive, active and passive user turns.	Figure 5c	Describes how assistant scaffolding supply varies with the user's current engagement state.
Prior-state support-form regression	Adjacent-turn logistic regression for whether the next user turn is constructive, adding prior-state $\times$ M1–M6 support-form interactions to the integrated model.	Figure 5d and Supplementary Table C13	Tests whether local coupling depends jointly on the user's immediately preceding engagement state and the form of scaffolded support.
Around-first-scaffold contrast	Mean constructive engagement after the first scaffolded assistant turn minus mean constructive engagement before that turn, among conversations containing scaffolded support.	Supplementary Figure D1	Exploratory boundary check only; the first scaffolded turn may be selected by difficulty, uncertainty or an already developing exchange.

Table C6 Uncertainty estimates for key support–engagement contrasts. Constructive-ratio contrasts are turn-weighted conversation-bootstrap estimates, with ratios computed as total constructive user turns divided by total user turns within scaffolded versus non-scaffolded conversation groups. Post-answer depth contrasts use conversation-level bootstrap resampling. Adjacent-turn contrasts bootstrap conversations over assistant-to-user pairs. Percentage-point and turn differences are the effect sizes reported in the main figures.

Setting	Contrast	Effect	95% CI	p value
WC coding	Has S2 − no S2 constructive ratio	+4.56 pp	[+4.15, +4.99]	< .001
WC coding	Has S2 − no S2 post-answer depth	+2.23 turns	[+2.11, +2.35]	< .001
LMSYS coding	Has S2 − no S2 constructive ratio	+2.51 pp	[+2.19, +2.86]	< .001
LMSYS coding	Has S2 − no S2 post-answer depth	+1.43 turns	[+1.33, +1.51]	< .001
SC coding	Has S2 − no S2 constructive ratio	+7.34 pp	[+5.29, +9.27]	< .001
SC coding	Has S2 − no S2 post-answer depth	+3.36 turns	[+2.69, +4.01]	< .001
WC writing	Has S2 − no S2 constructive ratio	+0.99 pp	[+0.82, +1.16]	< .001
WC writing	Has S2 − no S2 post-answer depth	+3.13 turns	[+2.96, +3.30]	< .001
LMSYS writing	Has S2 − no S2 constructive ratio	+1.09 pp	[+0.87, +1.31]	< .001
LMSYS writing	Has S2 − no S2 post-answer depth	+2.10 turns	[+1.91, +2.28]	< .001
SC writing	Has S2 − no S2 constructive ratio	+3.20 pp	[+1.55, +4.89]	< .001
SC writing	Has S2 − no S2 post-answer depth	+3.38 turns	[+1.92, +4.64]	< .001
WC coding	Adjacent S2 − reference lift	+2.68 pp	[+2.22, +3.09]	< .001
LMSYS coding	Adjacent S2 − reference lift	+2.01 pp	[+1.54, +2.55]	< .001
SC coding	Adjacent S2 − reference lift	+6.21 pp	[+4.35, +8.10]	< .001
WC writing	Adjacent S2 − reference lift	+1.13 pp	[+0.88, +1.41]	< .001
LMSYS writing	Adjacent S2 − reference lift	+0.27 pp	[-0.01, +0.56]	.061
SC writing	Adjacent S2 − reference lift	+3.35 pp	[+0.80, +6.62]	.025

Table C7 Offset-rate sensitivity for scaffolded-support Poisson models. The primary Fig. 3b Poisson model uses raw constructive-turn counts with conversation length entered as a covariate. As a rate sensitivity, this table refits the same setting-specific mean model with `offset(log(user_turns))`. Cells report the `has_S2` rate ratio with 95% confidence intervals and two-sided p values using quasi-Poisson scaled standard errors.

Setting	Offset rate ratio [95% CI], p value	Pearson dispersion	Conversations
WC coding	1.57 [1.47, 1.68], < .001	1.29	31,878
LMSYS coding	1.47 [1.38, 1.56], < .001	1.13	31,879
SC coding	1.58 [1.31, 1.91], < .001	1.29	2,481
WC writing	1.36 [1.26, 1.46], < .001	1.15	39,534
LMSYS writing	1.70 [1.50, 1.94], < .001	1.12	21,023
SC writing	1.65 [1.27, 2.15], < .001	1.32	1,539

Table C8 Conversation-level context model for constructive engagement. The outcome is whether a conversation contains at least one constructive user turn. The logistic regression includes user framing, task ecology, conversation-length bucket and dataset fixed effects. Cells report odds ratios with 95% confidence intervals and two-sided p values from conversation-robust standard errors. The reference category is unintentional framing, writing-oriented task, 2–3 user turns and WildChat.

Predictor	OR [95% CI], p value
Intentional framing	3.03 [2.93, 3.13], < .001
Coding task ecology	1.89 [1.71, 2.08], < .001
4–6 user turns	2.07 [1.99, 2.15], < .001
7+ user turns	3.58 [3.42, 3.74], < .001
LMSYS Chat	0.70 [0.68, 0.73], < .001
ShareChat	2.29 [2.12, 2.46], < .001
Conversations	128,334

Table C9 Conversation-level constructive-rate sensitivity model. The outcome is the number of constructive user turns out of total user turns in each conversation, modelled with a grouped-binomial logistic regression. The model includes user framing, task ecology, conversation-length bucket and dataset fixed effects, using user-turn counts as denominators to reduce the mechanical opportunity effect in any-constructive-turn outcomes. Cells report odds ratios with 95% confidence intervals and two-sided p values from conversation-robust standard errors. The reference category is unintentional framing, writing-oriented task, 2–3 user turns and WildChat.

Predictor	OR [95% CI], p value
Intentional framing	2.78 [2.68, 2.89], < .001
Coding task ecology	1.67 [1.48, 1.88], < .001
4–6 user turns	1.13 [1.09, 1.18], < .001
7+ user turns	1.06 [1.01, 1.11], .011
LMSYS Chat	0.72 [0.69, 0.75], < .001
ShareChat	2.31 [2.13, 2.50], < .001
Conversations	128,334
User turns	490,932
Constructive user turns	24,181

Table C10 Setting-level integrated adjacent-turn models. Each column reports a separate within-setting logistic regression for whether the next user turn is constructive. The scaffolded-support row comes from a broad S2-only model. Support-form rows come from a decomposed model that includes broad S2 and co-occurring M1–M6 forms, so form coefficients compare variation within scaffolded support rather than replacing the broad S2 contrast. All models include prior user state, intentional framing, assistant-turn index and recoverable model/source fixed effects where available. Cells report odds ratios with 95% confidence intervals and two-sided p values from conversation-cluster robust standard errors.

Predictor	WC coding	LMSYS coding	SC coding	WC writing	LMSYS writing	SC writing
A2U pairs	92,042	71,411	9,001	124,214	55,580	5,825
Conversations	31,809	31,375	2,474	39,455	20,936	1,537
Model/source FE count	13	23	4	13	23	3
Scaffolded support	1.13 [1.08, 1.19] $< .001$	1.25 [1.16, 1.34] $< .001$	1.24 [1.09, 1.42] .001	1.57 [1.38, 1.78] $< .001$	1.04 [0.86, 1.26] .679	1.34 [0.98, 1.85] .068
Prior user constructive	5.20 [4.84, 5.59] $< .001$	5.98 [5.40, 6.63] $< .001$	4.13 [3.46, 4.93] $< .001$	8.28 [6.73, 10.19] $< .001$	11.89 [9.10, 15.54] $< .001$	6.23 [4.43, 8.76] $< .001$
Prior user active	1.67 [1.57, 1.76] $< .001$	1.57 [1.45, 1.70] $< .001$	1.56 [1.32, 1.83] $< .001$	1.91 [1.68, 2.18] $< .001$	2.18 [1.80, 2.64] $< .001$	1.47 [1.07, 2.01] .017
Prior user passive	2.19 [1.65, 2.92] $< .001$	1.96 [1.34, 2.86] $< .001$	1.82 [1.25, 2.66] .002	2.49 [1.69, 3.65] $< .001$	2.32 [1.29, 4.17] .005	0.86 [0.40, 1.84] .703
Intentional framing	1.57 [1.49, 1.66] $< .001$	1.47 [1.36, 1.59] $< .001$	1.57 [1.34, 1.83] $< .001$	1.02 [0.89, 1.16] .798	0.86 [0.70, 1.06] .150	1.37 [0.93, 2.02] .107
M1 feedback	1.19 [1.07, 1.31] .001	1.22 [0.99, 1.51] .066	1.28 [1.03, 1.60] .025	0.69 [0.53, 0.91] .008	0.81 [0.51, 1.31] .397	1.74 [1.12, 2.71] .013
M2 hinting	1.04 [0.95, 1.14] .429	0.91 [0.74, 1.13] .410	0.85 [0.65, 1.10] .209	1.52 [1.20, 1.93] $< .001$	1.63 [1.09, 2.44] .018	0.76 [0.45, 1.29] .314
M3 instructing	0.89 [0.83, 0.96] .003	0.84 [0.68, 1.03] .102	0.93 [0.77, 1.13] .491	1.36 [1.04, 1.78] .027	1.06 [0.71, 1.58] .779	0.94 [0.45, 1.94] .860
M4 explaining	1.21 [1.09, 1.35] $< .001$	1.37 [1.09, 1.73] .007	1.77 [1.35, 2.33] $< .001$	1.81 [1.40, 2.34] $< .001$	1.12 [0.75, 1.68] .570	1.09 [0.58, 2.06] .789
M5 modelling	0.86 [0.80, 0.93] $< .001$	0.74 [0.63, 0.86] $< .001$	0.76 [0.60, 0.96] .021	0.69 [0.47, 1.01] .055	1.04 [0.64, 1.68] .872	1.57 [0.94, 2.61] .084
M6 questioning	0.73 [0.62, 0.86] $< .001$	0.31 [0.21, 0.45] $< .001$	0.70 [0.54, 0.90] .006	1.13 [0.74, 1.72] .567	0.66 [0.36, 1.21] .179	1.66 [0.90, 3.08] .106

Table C11 Integrated adjacent-turn regression combining user state, assistant scaffolding and model controls. The outcome is whether the next user turn is constructive. The scaffolded-support row is estimated in a broad S2-only model; M1, M4 and M6 rows are estimated in a decomposed support-form model that includes broad S2 and co-occurring support forms. Estimates are odds ratios with 95% confidence intervals and two-sided p values from conversation-cluster robust standard errors. The primary pooled model uses dataset fixed effects; the model/source sensitivity adds recoverable assistant model or source-family fixed effects.

Predictor	Pooled dataset FE	Pooled model/source FE	WildChat model FE
Scaffolded support	1.50 [1.44, 1.55], $< .001$	1.47 [1.42, 1.52], $< .001$	1.56 [1.49, 1.63], $< .001$
Prior user constructive	8.90 [8.45, 9.37], $< .001$	8.55 [8.11, 9.01], $< .001$	9.11 [8.53, 9.74], $< .001$
Prior user active	2.62 [2.52, 2.72], $< .001$	2.57 [2.47, 2.67], $< .001$	2.85 [2.71, 3.00], $< .001$
Prior user passive	2.07 [1.75, 2.43], $< .001$	2.06 [1.75, 2.43], $< .001$	2.32 [1.85, 2.90], $< .001$
Intentional framing	1.40 [1.35, 1.46], $< .001$	1.42 [1.37, 1.48], $< .001$	1.40 [1.33, 1.47], $< .001$
Coding task	1.37 [1.22, 1.54], $< .001$	1.38 [1.23, 1.55], $< .001$	1.21 [1.05, 1.40], .009
M1 feedback	0.79 [0.73, 0.86], $< .001$	0.80 [0.74, 0.86], $< .001$	0.74 [0.67, 0.82], $< .001$
M4 explaining	2.15 [1.99, 2.32], $< .001$	2.12 [1.96, 2.30], $< .001$	2.11 [1.92, 2.31], $< .001$
M6 questioning	0.82 [0.72, 0.93], .002	0.84 [0.74, 0.96], .008	1.08 [0.93, 1.25], .340
Model/source block	Reference	LR $\chi^2_{41} = 718.2$, $p < .001$	LR $\chi^2_{13} = 553.2$, $p < .001$

Table C12 Incremental contribution of scaffolding features in integrated adjacent-turn models. Likelihood-ratio block tests compare nested logistic regressions for whether the next user turn is constructive. All models include prior user state, user framing, task ecology, assistant-turn index and the indicated dataset or model/source fixed effects.

Scope	Added feature block	LR χ^2 (df)	p value
six task settings, dataset FE	Broad S2 after user/context controls	585.0 (1)	$< .001$
six task settings, dataset FE	M1–M6 descriptors within S2	895.1 (6)	$< .001$
six task settings, dataset FE	Full scaffolding block after user/context controls	1480.1 (7)	$< .001$
six task settings, model/source FE	Broad S2 after user/context controls	509.5 (1)	$< .001$
six task settings, model/source FE	M1–M6 descriptors within S2	827.9 (6)	$< .001$
six task settings, model/source FE	Full scaffolding block after user/context controls	1337.5 (7)	$< .001$
WildChat only, model FE	Broad S2 after user/context controls	418.6 (1)	$< .001$
WildChat only, model FE	M1–M6 descriptors within S2	519.4 (6)	$< .001$
WildChat only, model FE	Full scaffolding block after user/context controls	937.9 (7)	$< .001$

Table C13 Prior user state and support form jointly structure adjacent-turn constructive engagement. The first panel reports likelihood-ratio tests adding prior-state × M1–M6 interactions to the integrated adjacent-turn logistic model. The second panel reports pooled model/source fixed-effect estimates for selected support forms within each prior user state. Estimates are odds ratios with 95% confidence intervals and two-sided p values from conversation-cluster robust standard errors.

Panel	Scope / prior state	Term or test	Estimate
Interaction block	six task settings, dataset FE	Prior state × M1–M6	LR $\chi^2_{18} = 308.6$, $< .001$
Interaction block	six task settings, model/source FE	Prior state × M1–M6	LR $\chi^2_{18} = 300.2$, $< .001$
Interaction block	WildChat only, model FE	Prior state × M1–M6	LR $\chi^2_{18} = 186.0$, $< .001$
State-stratified model/source FE	prior constructive	M1 feedback	0.72 [0.65, 0.81], $< .001$
State-stratified model/source FE	prior constructive	M4 explaining	1.72 [1.45, 2.04], $< .001$
State-stratified model/source FE	prior active	M1 feedback	0.83 [0.73, 0.95], .008
State-stratified model/source FE	prior active	M4 explaining	2.12 [1.87, 2.40], $< .001$
State-stratified model/source FE	prior passive	M1 feedback	2.16 [1.16, 4.04], .016
State-stratified model/source FE	prior passive	M4 explaining	2.88 [1.55, 5.34], $< .001$

Table C14 Support-supply profile within scaffolded assistant turns. Values show the percentage of scaffolded assistant turns containing each scaffolding-means label. Means labels are not mutually exclusive, so rows do not sum to 100%. Coding-oriented scaffolded turns are dominated by explanation-heavy support, whereas writing-oriented scaffolded turns show a more mixed support profile.

Setting	M1 Feedback	M2 Hinting	M3 Instructing	M4 Explaining	M5 Modelling	M6 Questioning
WildChat coding	8.8	14.7	29.7	84.0	23.9	6.9
LMSYS coding	4.5	8.2	10.4	79.2	18.3	10.6
ShareChat coding	13.3	12.7	25.7	81.2	18.4	18.6
WildChat writing	16.6	9.6	59.2	29.5	15.0	6.9
LMSYS writing	11.3	10.9	52.3	26.9	10.3	13.4
ShareChat writing	24.2	23.2	22.9	45.2	23.5	38.1

Table C15 Boundary-check corpora processed with the common pipeline. SWE-chat and ThoughtTrace were labelled and analysed with the same Level 1–4 workflow, but are not part of the main six-setting analysis because they do not provide stable coding–writing task comparisons or stable non-scaffolded reference variation. Percentages for cognitive, constructive and emotional engagement use user turns as the denominator; scaffolded-support percentages use assistant turns as the denominator.

Corpus / setting	Use in manuscript	Conv.	User turns	Assistant turns	Scaffolded turns (%)	Cognitive overall (%)	Constructive level (%)	Boundary-check implication
SWE-chat coding	Coding-only appendix check	1,536	9,506	6,813	19.45	32.36	17.96	Scaffolded conversations were more constructive than non-scaffolded conversations (0.173 vs. 0.133) and deeper after the first answer (+6.41 turns), but the corpus lacks a usable writing counterpart.
SWE-chat writing	Excluded from inference	3	18	14	42.86	22.22	11.11	Only three conversations remained after filtering, so estimates are not interpretable.
ThoughtTrace coding	Feasibility check only	56	251	252	85.71	64.54	9.96	All conversations contained scaffolded support, leaving no non-scaffolded conversation-level reference group for support-association estimates.
ThoughtTrace writing	Feasibility check only	61	301	300	67.33	51.16	2.66	Scaffolded support remained very common and only two conversations lacked scaffolded support, making adjusted support contrasts unstable.

Appendix D Supporting figures

The supporting figures report extended temporal checks and model-snapshot robustness analyses where model metadata are available. They are included to make the main claims auditable without shifting the manuscript's emphasis away from the six main task settings. Model-snapshot figures are descriptive robustness checks because model labels in naturalistic corpora are observational rather than experimentally assigned.

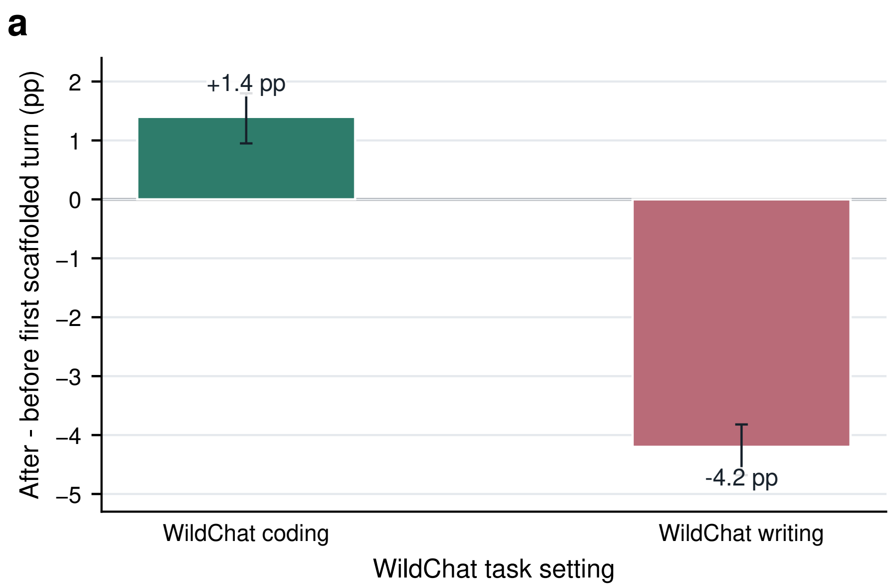

Fig. D1 Coarse around-first-scaffold contrasts are exploratory boundary checks. Bars show the difference in constructive engagement after versus before the first scaffolded assistant turn within the WildChat robustness subset of conversations containing scaffolded support. These before–after contrasts are not used as treatment-effect or accumulation estimates because the first scaffolded turn may be selected by difficulty, uncertainty or an already developing exchange.

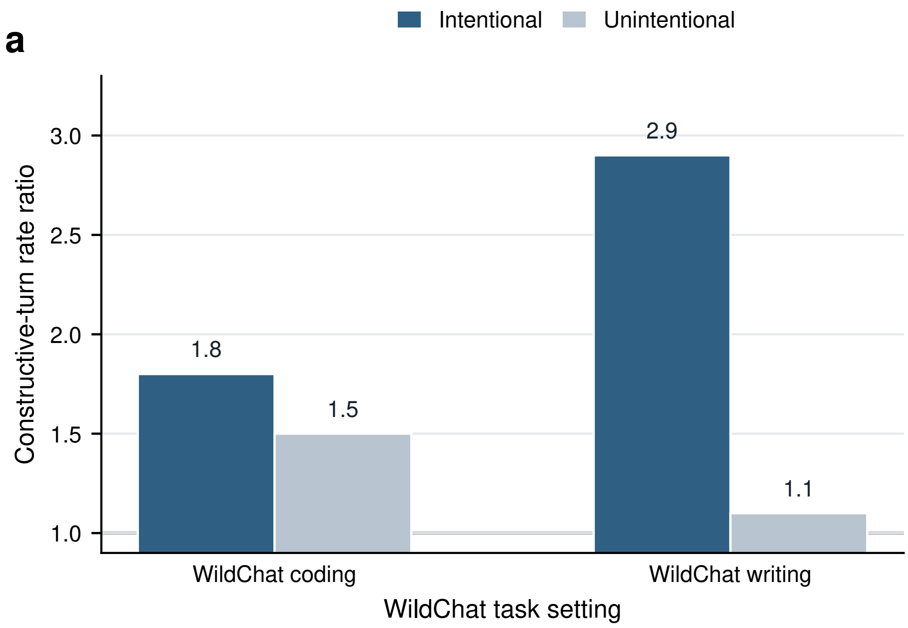

Fig. D2 Scaffolded-support associations vary by user framing. Stratified model estimates compare the association between scaffolded-support presence and constructive participation in intentional versus unintentional conversations. The figure is interpreted as descriptive moderation rather than evidence that intent causally changes support effectiveness, because user framing, task complexity and persistence are not randomly assigned.

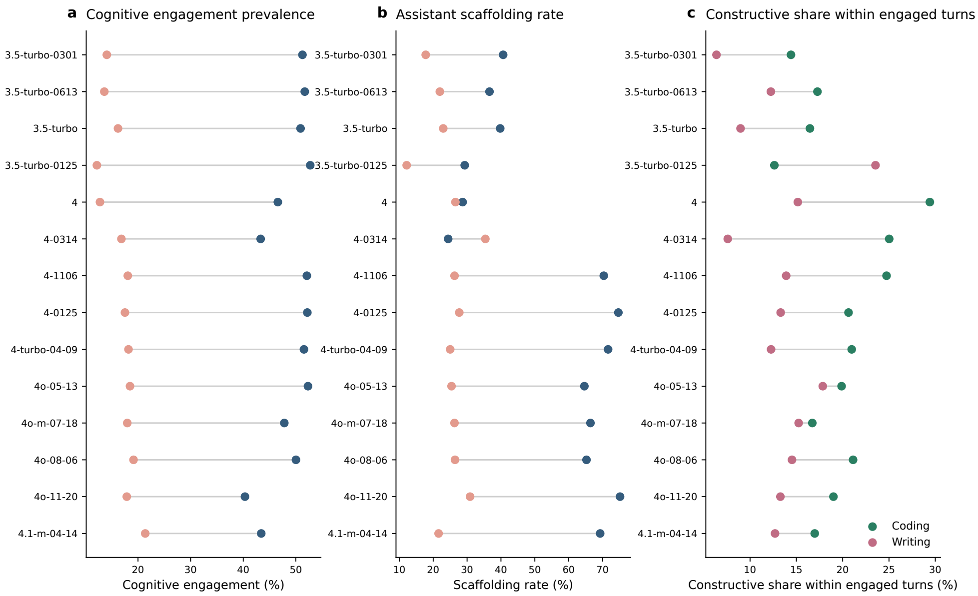

Fig. D3 WildChat model-snapshot prevalence profiles preserve the main domain contrast. Across user-facing WildChat model snapshots, coding-oriented conversations generally show higher cognitive engagement and higher assistant scaffolding rates than writing-oriented conversations. The model labels are observational and may reflect time, user mix and traffic composition, so the figure is used as a robustness check rather than a model-causal comparison.

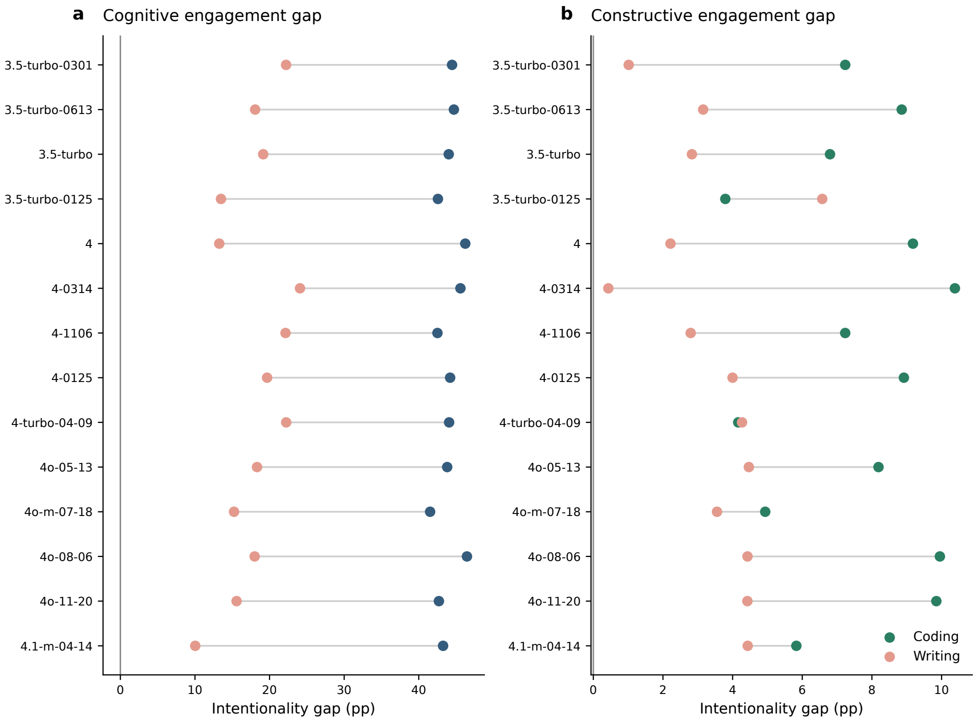

Fig. D4 Intentionality gaps are visible across WildChat model snapshots. The figure reports model-specific differences between intentional and unintentional conversations in cognitive and constructive engagement, separately for coding and writing. Most model snapshots preserve the directional pattern that intentionally framed conversations contain richer learning-oriented participation.

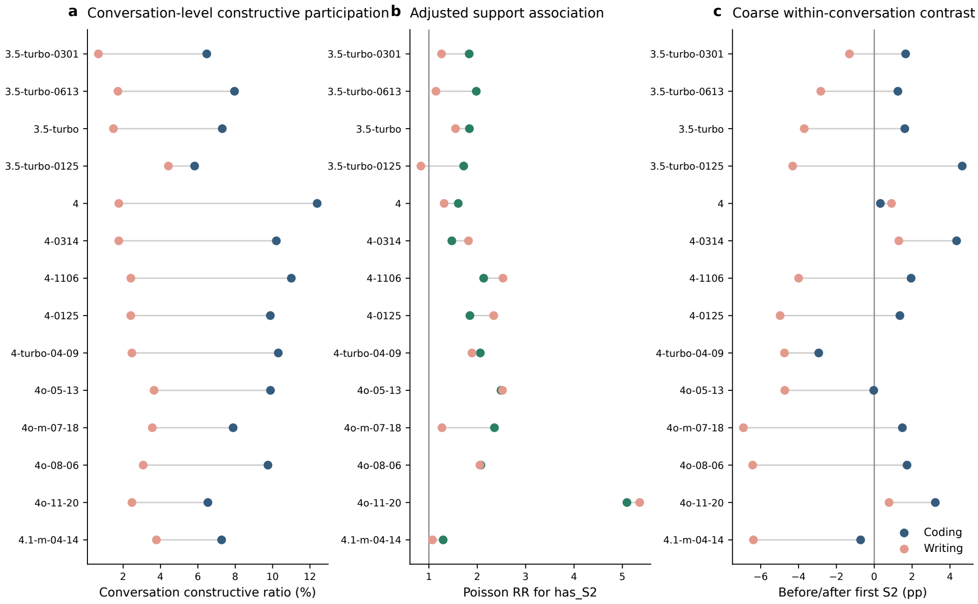

Fig. D5 Conversation-level support associations are broadly positive across WildChat model snapshots. The figure repeats major conversation-level analyses by user-facing WildChat model snapshot, including constructive participation, adjusted support associations and exploratory coarse before–after contrasts around the first scaffolded assistant turn. The pattern supports the distinction between stable conversation-level associations and more local adjacent-turn temporal claims.

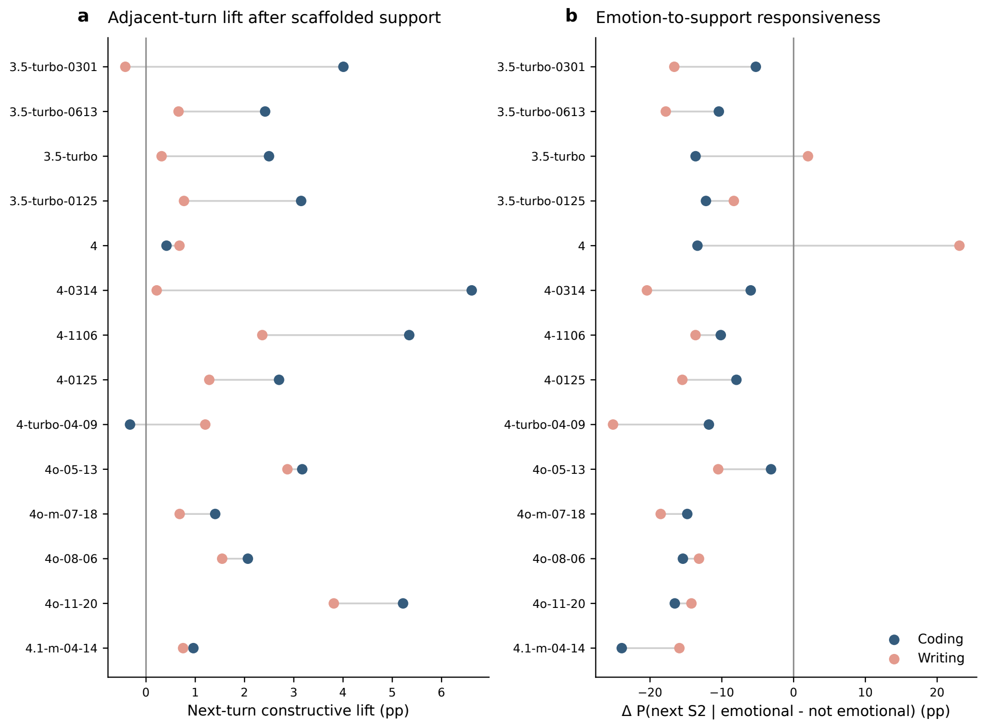

Fig. D6 Local temporal signals vary across WildChat model snapshots. The figure reports adjacent-turn constructive lifts and emotion-to-support responsiveness by model snapshot. Adjacent-turn constructive lifts are often positive, especially in coding-oriented conversations, but the magnitude varies across snapshots; emotion-to-support responsiveness is more heterogeneous and is treated as exploratory.

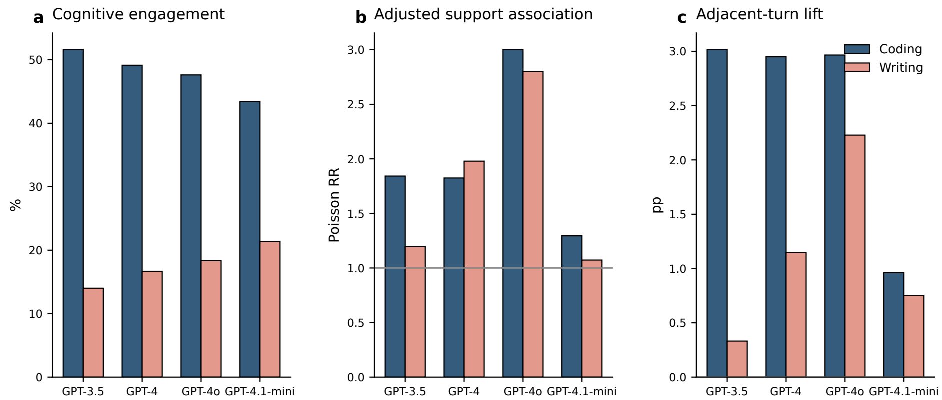

Fig. D7 Coarse model-family aggregation preserves the main directional patterns. WildChat model snapshots are grouped into coarse model families to summarize cognitive engagement prevalence, adjusted support associations and adjacent-turn constructive lift. Coding remains higher than writing across families, and the support-association and local-lift patterns remain directionally consistent on average.